tsitski ProBuilder Tools

UNITY ASSET | PROBUILDER 6 | UNITY 6.4+

A live boolean cutter, level generators and modelling tools for Unity ProBuilder.

Contents

GETTING STARTED

- Overview
- Installation
- Tools menu and popup
- Editing points and paths
- Stats assets

COMPONENTS

- ProBuilder Cutter
- ProBuilder Trim
- ProBuilder Array
- ProBuilder Stairs
- ProBuilder Pillar
- Walls From Floor Plan

MODELLING TOOLS

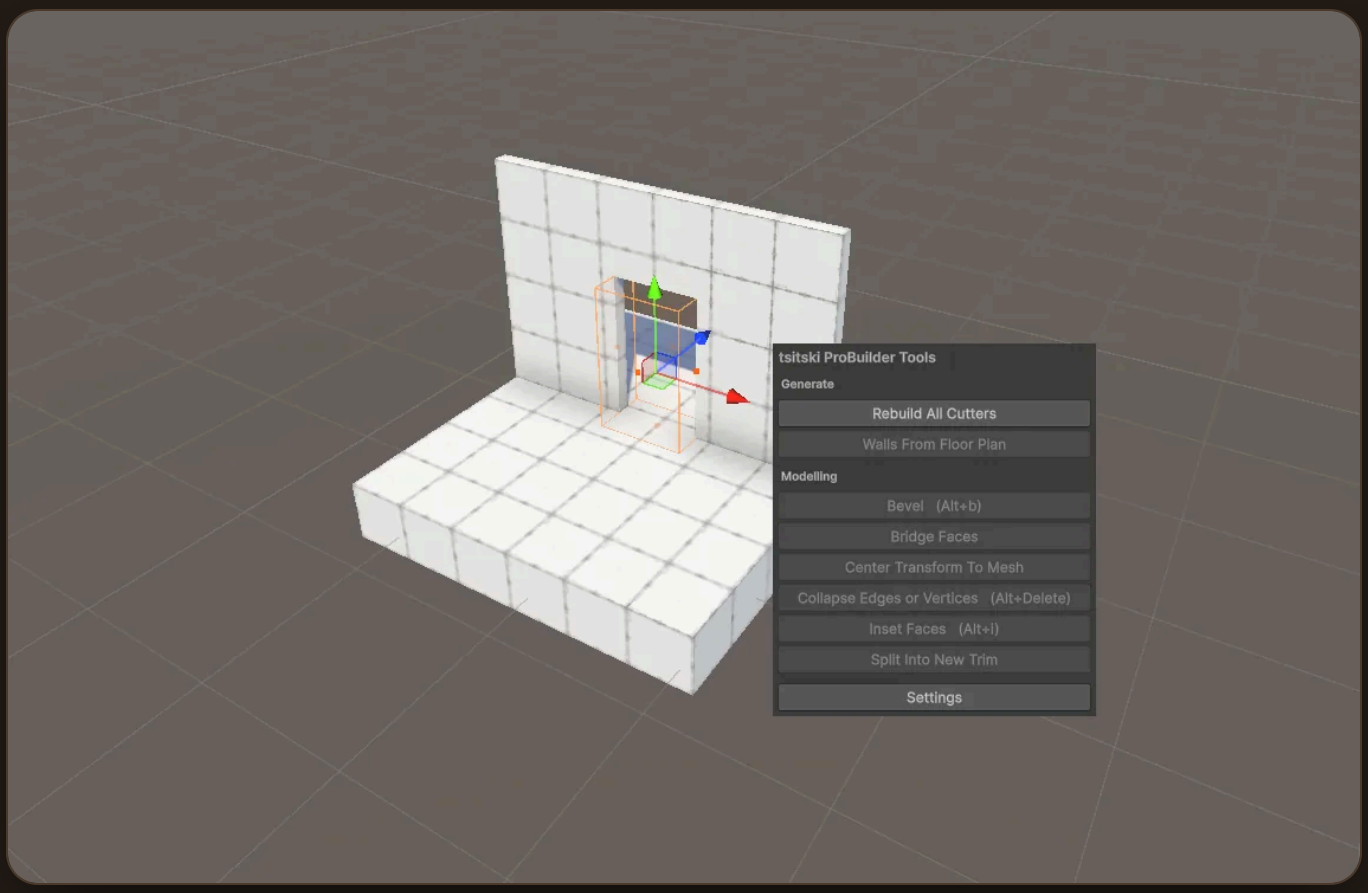
- Bevel and Inset Faces
- Bridge Faces
- Collapse Edges or Vertices
- Split Into New Trim
- Center Transform To Mesh

HELP

- Settings
- Troubleshooting
- Credits

Overview

tsitski ProBuilder Tools adds live, non-destructive building blocks for level design in Unity ProBuilder: a boolean cutter for doors and windows, generators for trims, arrays, stairs, pillars and walls, and a set of modelling tools that are one shortcut away in the Scene view.



Doorways and windows cut live into a ProBuilder wall.

Components

- **ProBuilder Cutter** — a box, rounded box, cylinder, polygon, explosion or path that cuts a hole through any ProBuilder mesh it overlaps, live in the editor.
- **ProBuilder Trim** — sweeps a 2D profile along a path for skirting, mouldings, pipes and railings.
- **ProBuilder Array** — copies a source mesh along a path, evenly spaced.
- **ProBuilder Stairs** — stairs along a path, with landings, winding corners, an optional stairwell cutter, side trims and a smooth collider.
- **ProBuilder Pillar** — a profile spun into a round or square pillar, with a base, a repeating or stretching shaft and a capital.

- **ProBuilder Floor Plan** — walls generated around the outline of a floor.

Modelling tools

- **Bevel** and **Inset Faces** — with a live settings window, like Blender's redo panel.
- **Bridge Faces** — join two groups of faces with a tube or tunnel.
- **Collapse Edges or Vertices** — merge flat faces and clean up leftover vertices.
- **Split Into New Trim** and **Center Transform To Mesh**.
- **Tools popup** — press `Alt` + `X` in the Scene view to get every tool under your cursor.

Installation

1. Buy or add the package on the Asset Store, then open `Window` › `Package Manager`, choose **My Assets**, and import **tsitski ProBuilder Tools**.
2. ProBuilder is installed automatically as a dependency if you don't have it yet.
3. Everything lands in `Assets/tsitski/tsitski ProBuilder Tools`. Open the demo scene in its `Demo` folder to see each tool in action.

NOTE

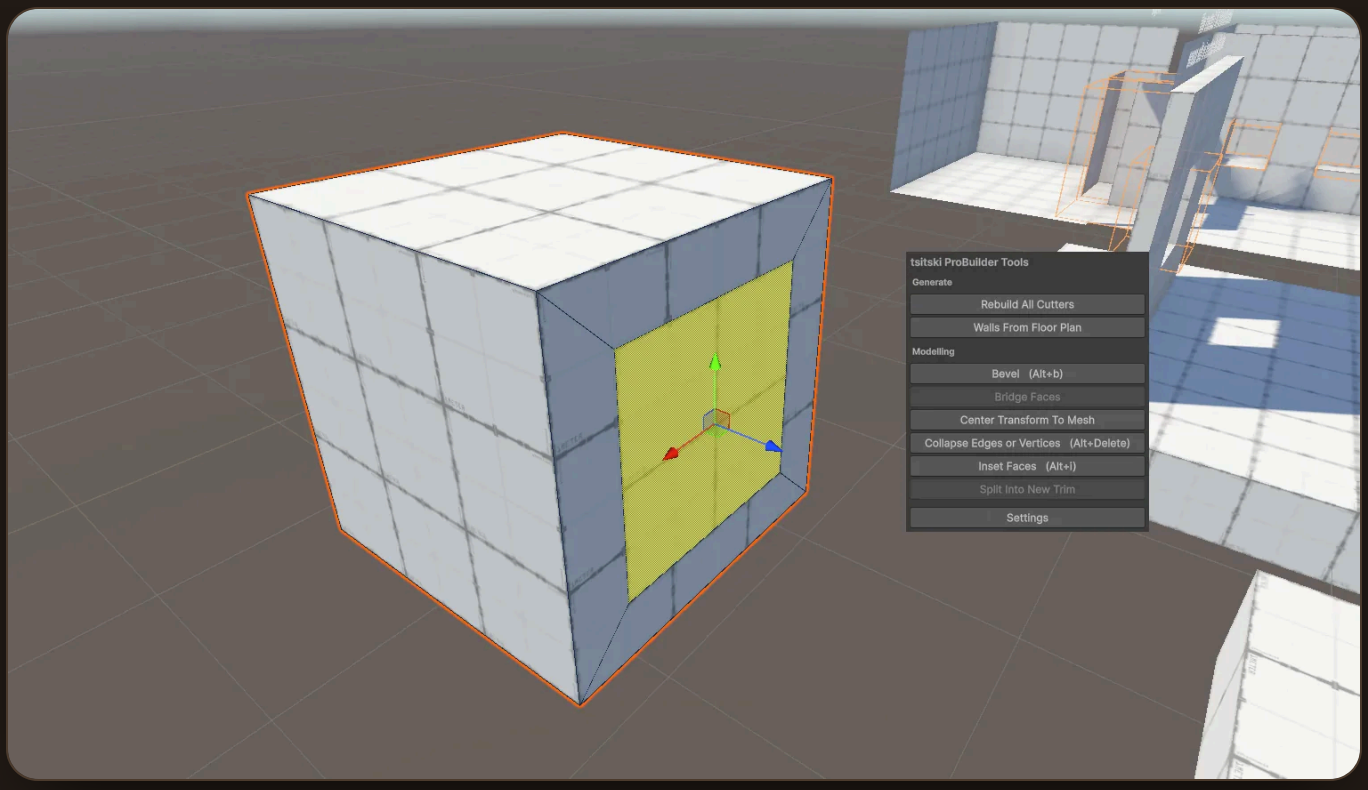
Requires Unity 6.4 (6000.4) or newer and ProBuilder 6. Works with the Built-in, URP and HDRP render pipelines.

TIP

If the demo scene's text shows up pink or Unity asks to import **TMP Essentials**, click **Import TMP Essentials**. It's a one-time setup for TextMeshPro.

Tools menu and popup

Every tool lives in `Tools` › `tsitski ProBuilder Tools`, split into **Modelling** and **Generate**, with **Settings** at the bottom. Faster: hover the Scene view and press `Alt` + `X` to open a popup with all of them, grouped the same way, right under your cursor. Tools that can't run on your current selection are greyed out.



Alt + X opens every tool at your cursor.

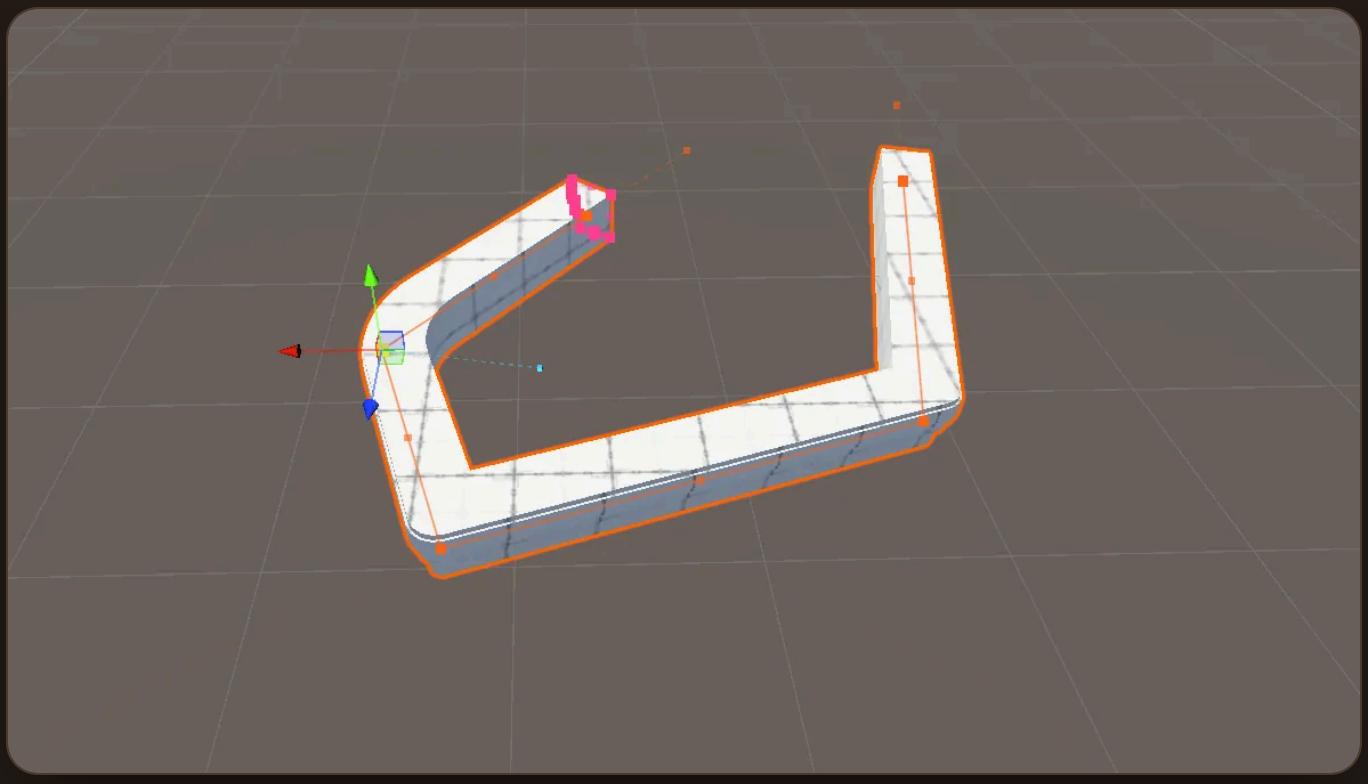
PROPERTY	SHORTCUT	WHAT IT DOES
Tools popup	Alt + X	Open every tool at the cursor.
Bevel	Alt + B	Bevel the selected edges or faces.
Inset Faces	Alt + I	Inset the selected faces.
Collapse	Alt + Delete	Collapse the selected faces, edges or vertices.

TIP

Change any shortcut in `Edit` › `Shortcuts`. The popup is under **tsitski ProBuilder Tools › Tools Popup**.

Editing points and paths

Cutters, trims, arrays, stairs and pillars all edit their points the same way, right in the Scene view. Select the object and its points appear.



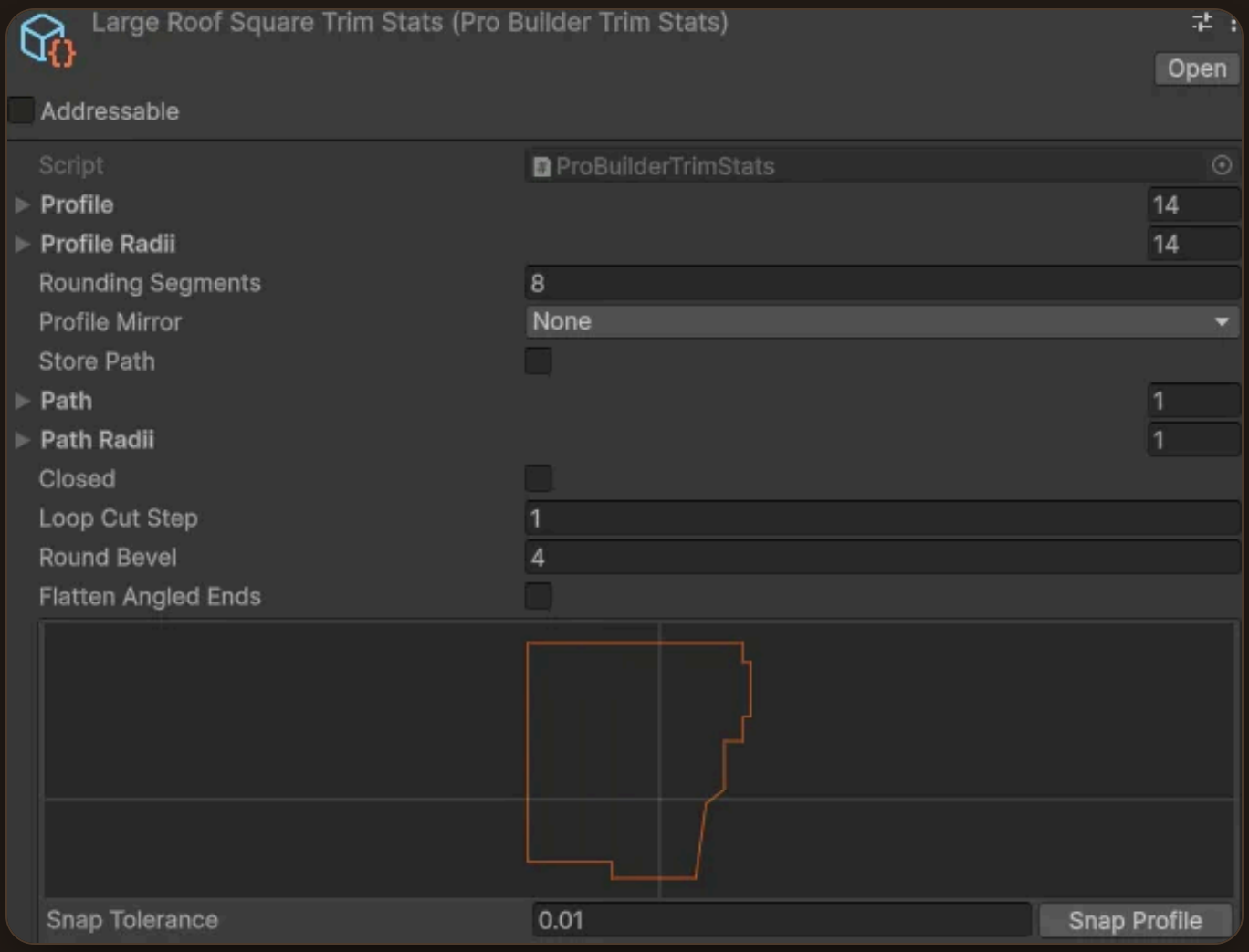
Points, edge-midpoint dots, extend dots and the cyan rounding handle.

- Click a point to select it. `Shift` + click adds or removes points, and `Shift` + drag on empty space box-selects them. `Shift` + click on another object still adds it to the selection as normal.
- Selected points follow the current tool: Move (`W`), Rotate (`E`) or Scale (`R`), around the middle of the selection. On a Path cutter, rotating also tilts the cutter's box at each selected point, so even a single point can be rotated.
- Click the small dot in the middle of an edge to insert a point there.
- On an open path, click the dot past either end to extend the path.
- Drag the cyan handle on a selected corner to round it. Rounding is saved per point, and the object's **Rounding Segments** (or **Corner Segments**) sets how smooth it is.
- `Delete` removes the selected points and `Esc` clears the selection.
- `Ctrl` + `C` and `Ctrl` + `V` (`Cmd` on Mac) copy and paste points, even between objects. Pasted points start right after the copied ones; as you drag them they swap to before the copied point if that keeps the path tidier.

Points are edited only when exactly one object is selected. Generated meshes are rebuilt from their settings, so ProBuilder edits made directly to them are overwritten.

Stats assets

Cutters, trims, arrays, stairs and pillars each have a matching **Stats** asset (create one with `Assets` › `Create` › `tsitski ProBuilder Tools` or the inspector's **Save As … Stats** button). Assign it to share settings between many objects: editing the asset updates every object that uses it.



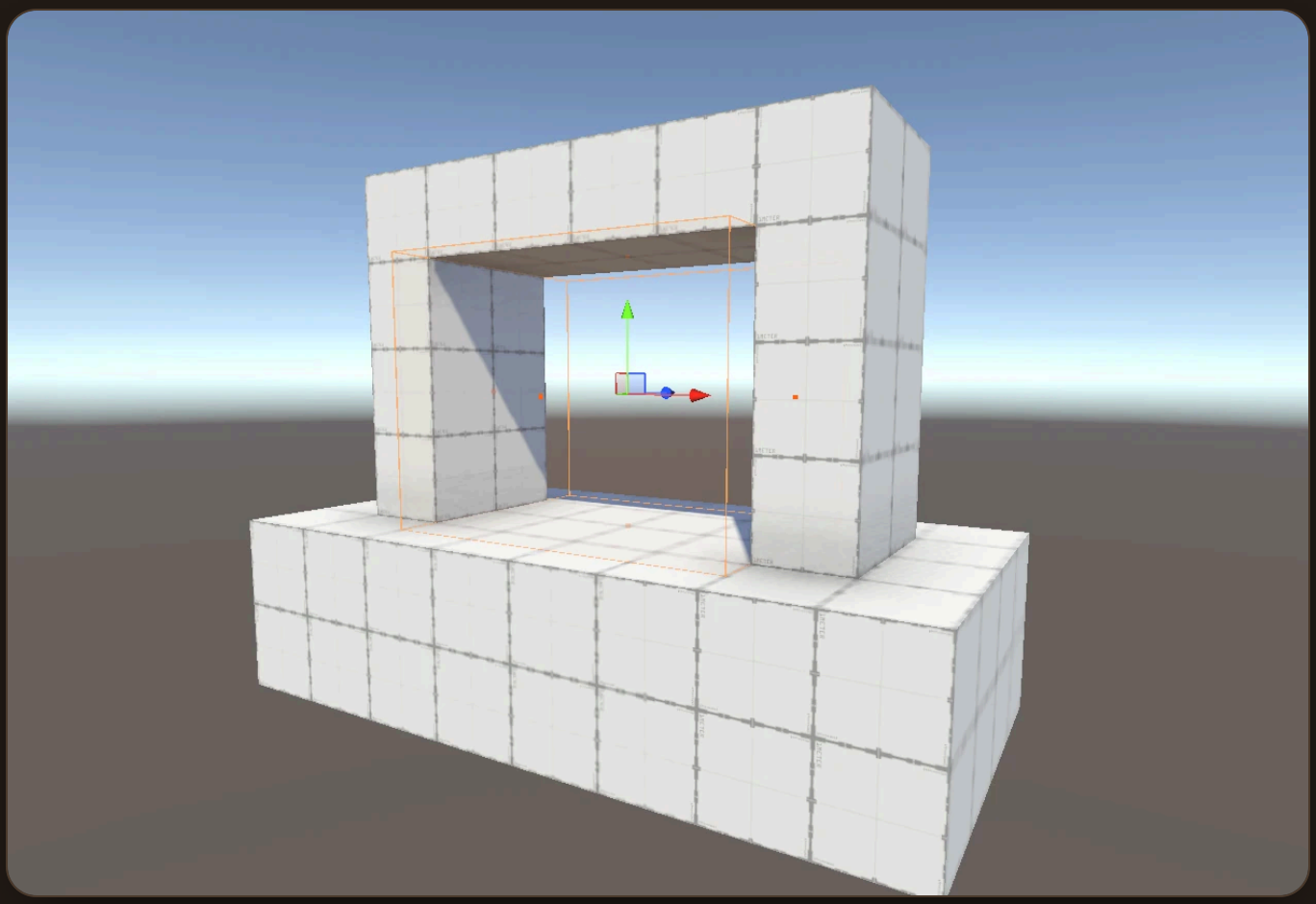
Settings held by the Stats asset are hidden on the component.

- While a Stats asset is assigned, the settings it holds are hidden on the component.
- **Copy Stats Into …** copies the asset's settings back onto the component and unassigns it, so you can tweak one object on its own.
- Every inspector also has a **Rebuild** button.

> TIP
> Hover any field in an inspector for a tooltip explaining it.

ProBuilder Cutter

A ProBuilder Cutter is a solid shape. Wherever it overlaps a ProBuilder mesh, that part of the mesh is cut away. The cut updates live as you move, scale or edit the cutter or the mesh, so you can slide a doorway along a wall and watch it follow.



The orange box is the cutter; the hole follows it.

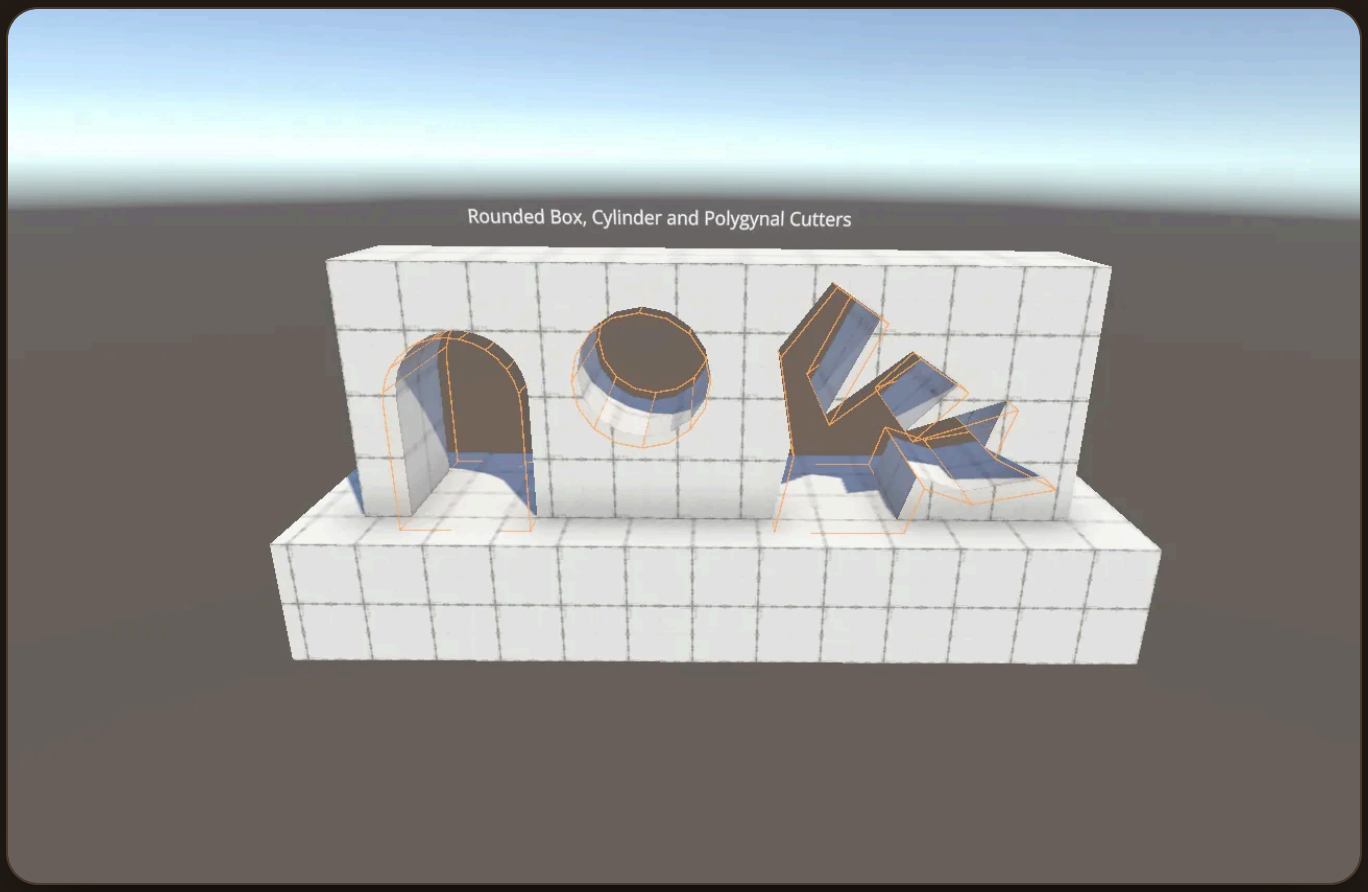
Quick start

1. Create an empty GameObject and place it where you want the hole.
2. Add the component from `Add Component` › `tsitski ProBuilder Tools` › `ProBuilder Cutter`.
3. Set its **Size** so the box pokes all the way through the wall, then move it into place. The hole appears as soon as the box goes into the mesh.

Shapes

Pick the cutter's shape with **Shape**. Every shape except Path is an outline drawn on the cutter's local XY plane and pushed through along its local Z axis, so with no rotation it cuts

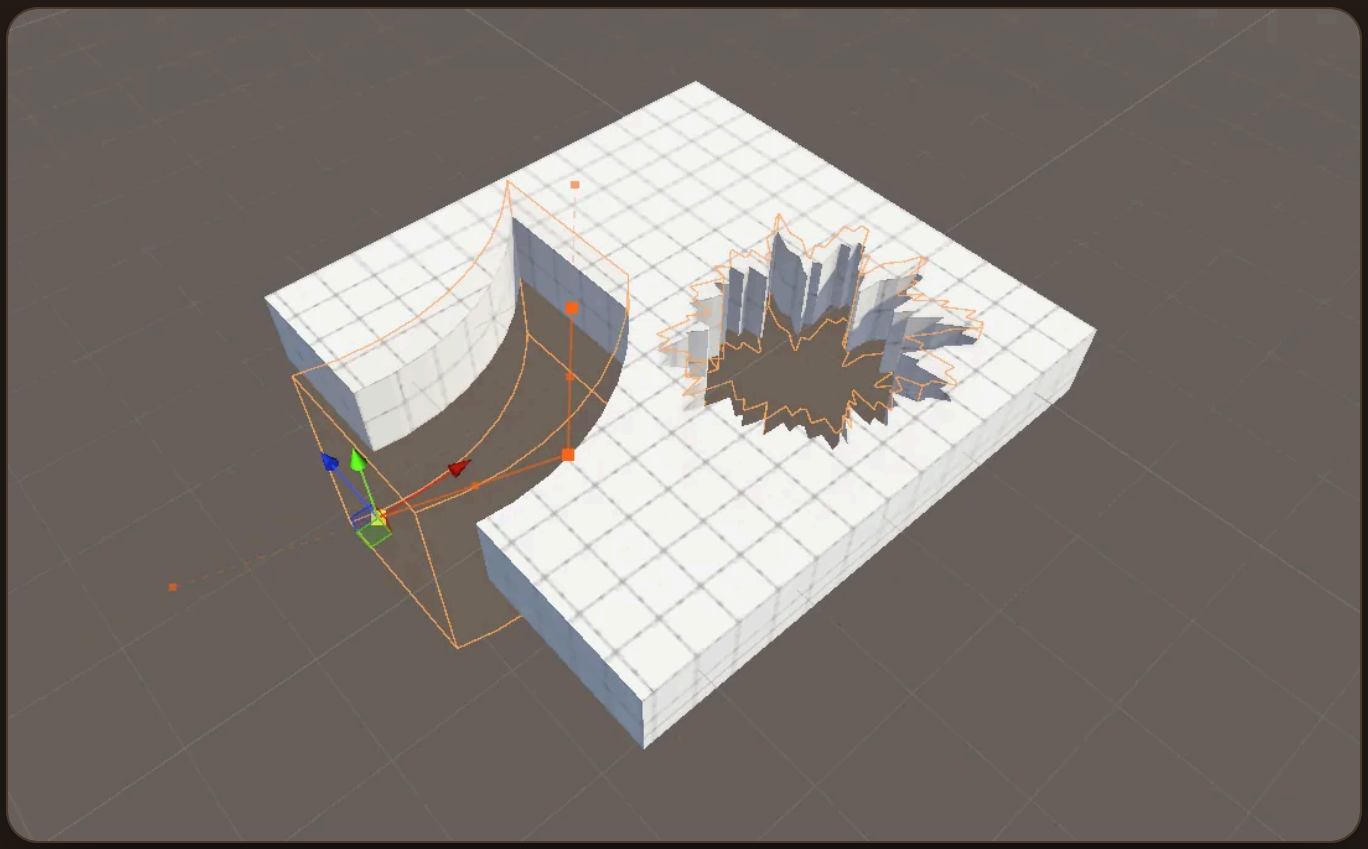
straight into a wall facing Z.



A rounded box, a cylinder and a polygon cutter.

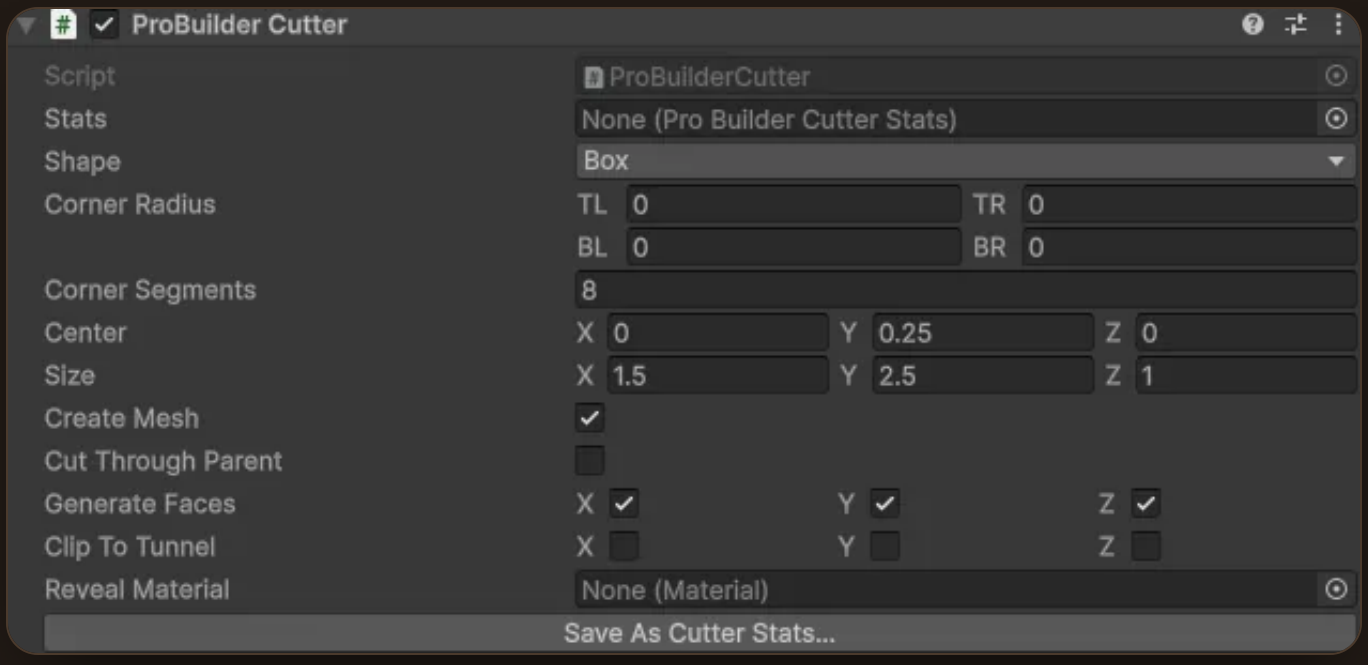
- **Box** — the default. Give any of its four corners a **Corner Radius** to round it. Round just the top two for an arched doorway.
- **Cylinder** — a round hole. A non-square size gives an oval.
- **Polygon** — any outline you like, including concave ones, edited in the Scene view. Its corners can be rounded too.
- **Path** — a box **Path Width** wide and **Path Height** tall, swept along a path of points in the cutter's local space. Corners can be rounded, and each point can be rotated with the Rotate tool to tilt or roll the box. Great for corridors, trenches and stairwells.
- **Explosion** — a jagged blast hole. Change the seed, spikes and jaggedness, or press **Randomize Explosion**.

Polygon and Path points are edited in the Scene view; see Editing points and paths.



A Path cutter and an Explosion cutter.

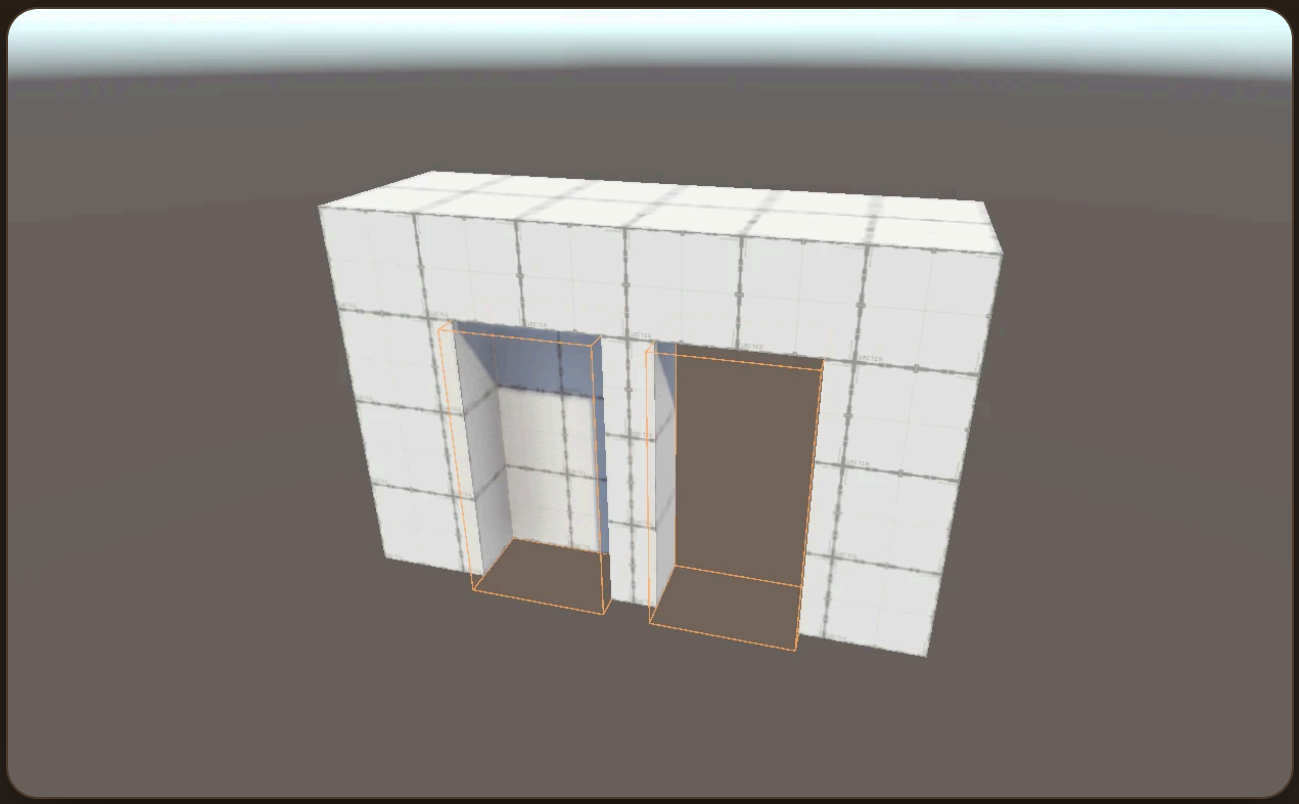
Properties



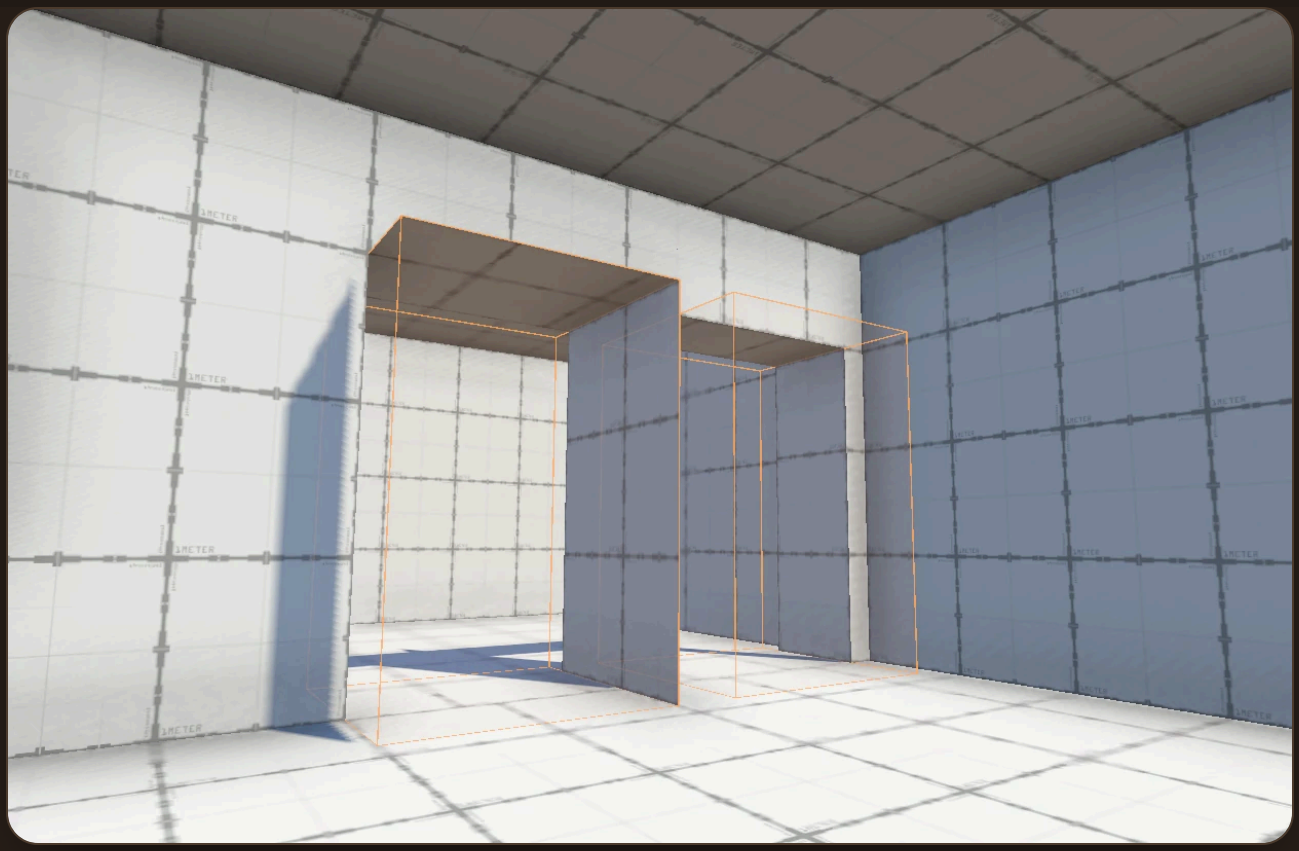
PROPERTY	TYPE	WHAT IT DOES
Stats	Cutter Stats	Shared settings for every cutter using the asset, including the shape, size and points. While it's assigned only this field is shown, and Scene view edits on any of those cutters edit the asset. **Copy Stats Into Cutter** unassigns it and keeps the settings.
Shape	enum	Box, Cylinder, Polygon, Path or Explosion. See Shapes above.
Corner Radius	TL / TR / BL / BR	Box only. Rounds each corner, in local units, up to half the box's shorter side.
Corner Segments	int	How many straight pieces each rounded corner is made of (Box, Polygon and Path).
Cylinder Segments	int	Cylinder only. How many sides the cylinder has.
Explosion settings	int, 0–1	Explosion only. Seed, Spikes (12), Minor Spikes (3) and Jaggedness (0.6) shape the blast hole.
Polygon Points	List of Vector2	Polygon only. The outline's points. Easiest to edit in the Scene view.
Path Points	List of Vector3	Path only. The points the box is swept along, centred on them.
Path Rotations	List of Quaternion	Path only. Rotation of the box at each point. Set it with the Rotate tool in the Scene view.
Path Width / Path Height	float	Path only. Size of the swept box.
Center / Size	Vector3	All shapes except Path. The box in local space. A cutter with any size of 0 is ignored.
Create Mesh	bool	Builds the inside walls of the hole. Turn it off to only remove the geometry inside the cutter.
Cut Through Parent	bool	A cutter that's a child of a ProBuilder mesh doesn't cut that mesh, or the other meshes under it, unless this is on. That lets a mesh carry its own cutter without cutting itself.

PROPERTY	TYPE	WHAT IT DOES
Generate Faces	X / Y / Z	Which sides of the cutter can become inside walls, per local axis. On a Path cutter, X is the sides, Y the top and bottom and Z the ends.
Clip To Tunnel	X / Y / Z	On that axis, ends the hole's walls at the mesh's own surface instead of at the cutter, so openings finish flush with sloped or uneven walls.
Reveal Material	Material	Material for the new inside walls. Leave it empty to use the mesh's first material.

▾ Generate Faces and Clip To Tunnel, side by side



Generate Faces: each axis toggles a pair of inside walls.

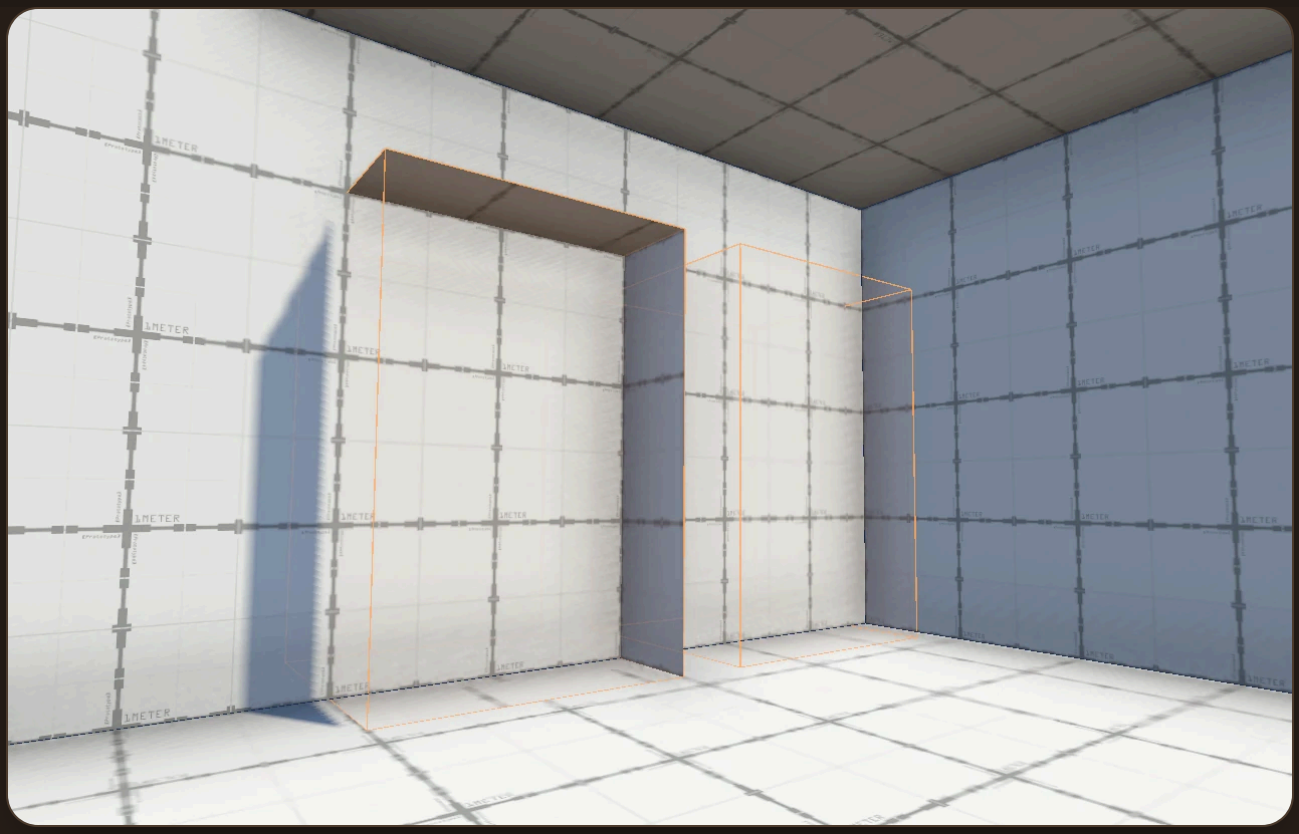


Clip To Tunnel: walls end at the wall's surface instead of the box.

Editing a mesh that has been cut

The cutter never changes your original ProBuilder mesh. The cut result is a separate mesh, created as a child of the original, and shown in its place.

When you select the mesh and switch to a ProBuilder editing mode (vertex, edge or face), the original uncut mesh comes back so you can edit it normally. Switch back to object mode and the cut result returns, rebuilt with your changes.



In ProBuilder's edit modes you edit the original; in object mode you see the cut.

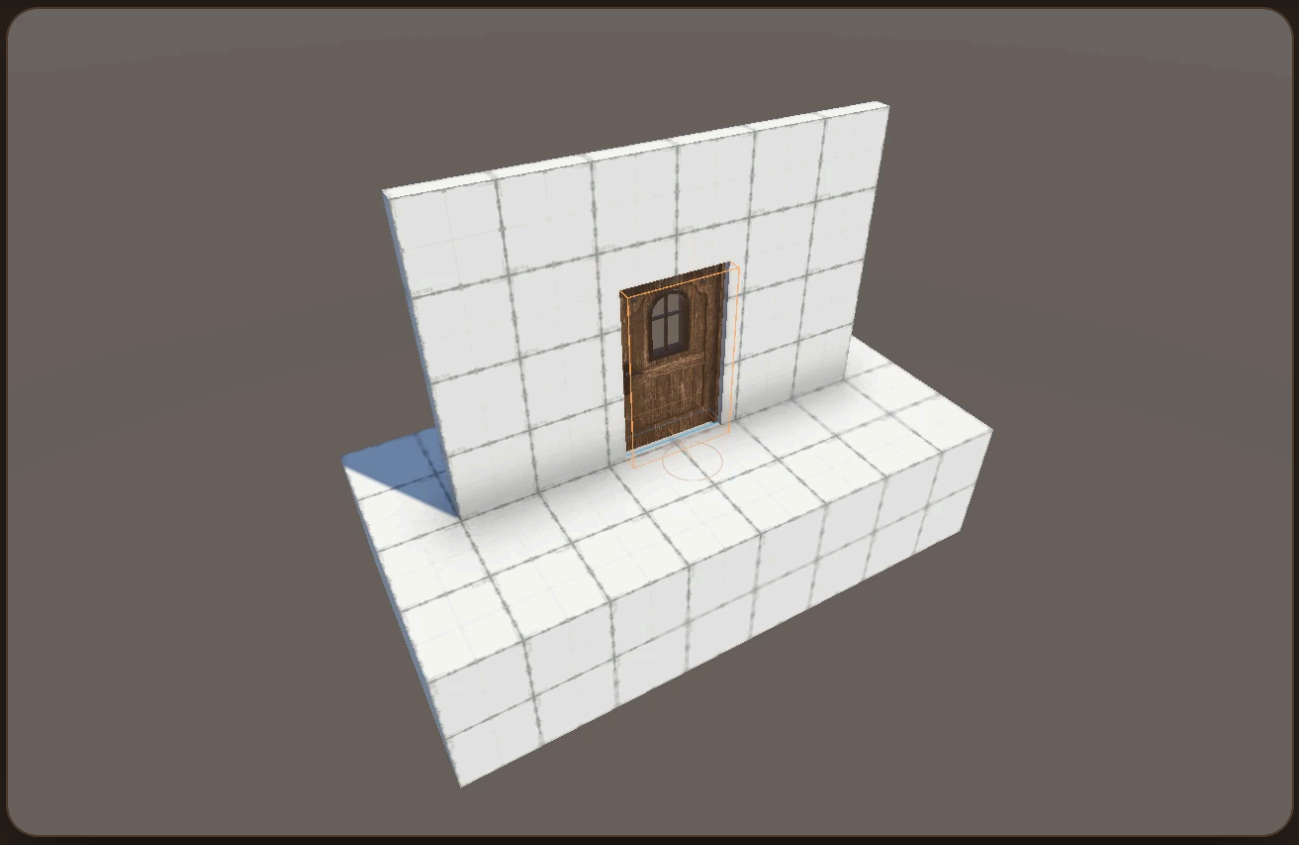
A **ProBuilder Cut Target** component is added to cut meshes automatically to keep track of this, and removed again when nothing cuts them. You never need to add or edit it yourself.

Cutting more than one mesh at once

A cutter cuts every ProBuilder mesh in the same scene that it overlaps. When it overlaps several, only the mesh it goes into most deeply gets the inside walls, so you don't end up with doubled walls.

Cutters inside prefabs (doors and windows)

A cutter never cuts meshes that belong to its own prefab or its own children. Put a cutter inside a door or window prefab, drop the prefab against any ProBuilder wall, and it cuts its own opening without cutting the door's own frame.



A window prefab carrying its own cutter.

Colliders, play mode and builds

- The cut result gets a Mesh Collider matching the cut shape, and the original mesh's colliders are switched off while it's cut.
- Cuts are made in the editor and saved with the scene as normal meshes. Nothing runs in play mode or in builds.
- The cut result copies its renderer settings from the original mesh.

Rebuilding every cut

Cuts rebuild by themselves whenever something changes. If a cut ever looks out of date, use `Tools` › `tsitski ProBuilder Tools` › `Generate` › `Rebuild All Cutters`.

Scripting

Everything is in the `Tsitski.ProBuilderTools` namespace. To resize a box cutter from an editor script and have the cut rebuild, use `SetBox`:

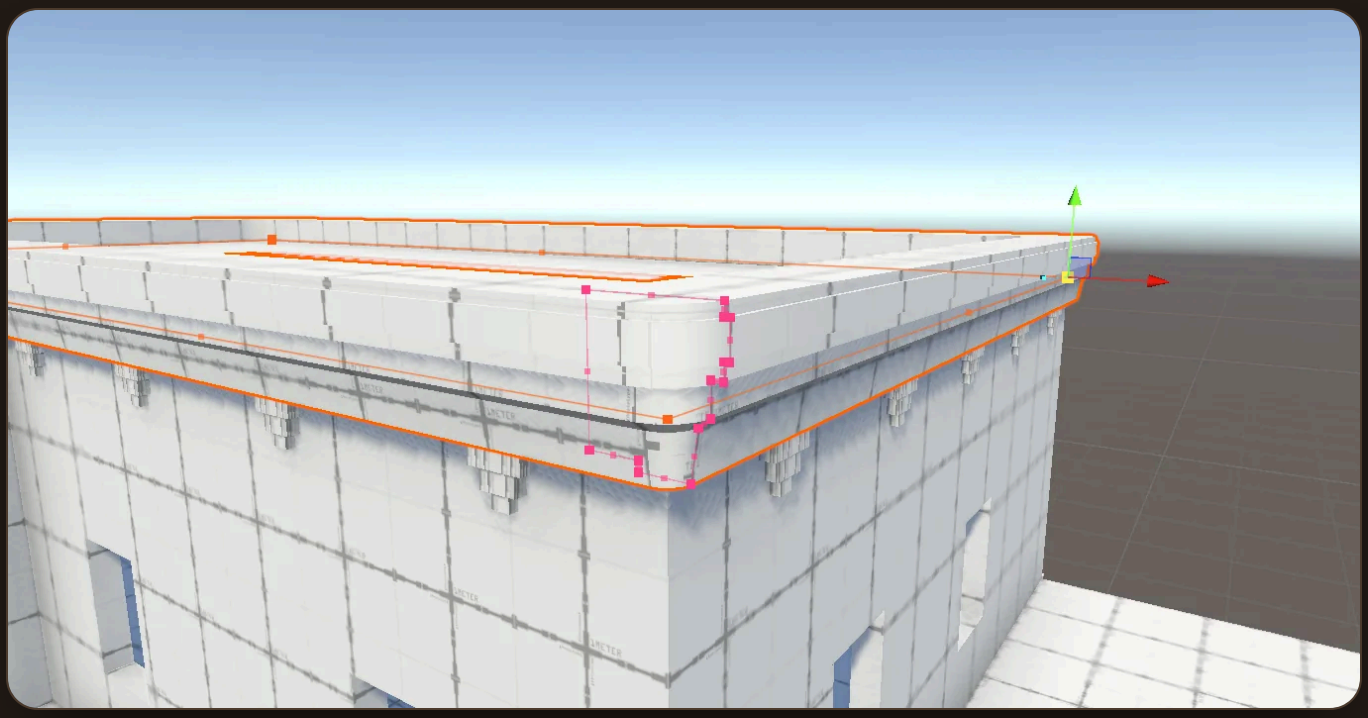
```
using Tsitski.ProBuilderTools;

cutter.SetBox(new Vector3(0f, 1f, 0f), new Vector3(1.2f, 2.1f, 0.6f));
```

After changing other fields from a script, call `cutter.NotifyChanged()`. Editing in the inspector, or moving, rotating or scaling the Transform, rebuilds automatically.

ProBuilder Trim

A trim sweeps a 2D profile along a path: skirting boards, crown mouldings, pipes, kerbs, railings. Both the path and the profile are edited in the Scene view, and the mesh rebuilds live.

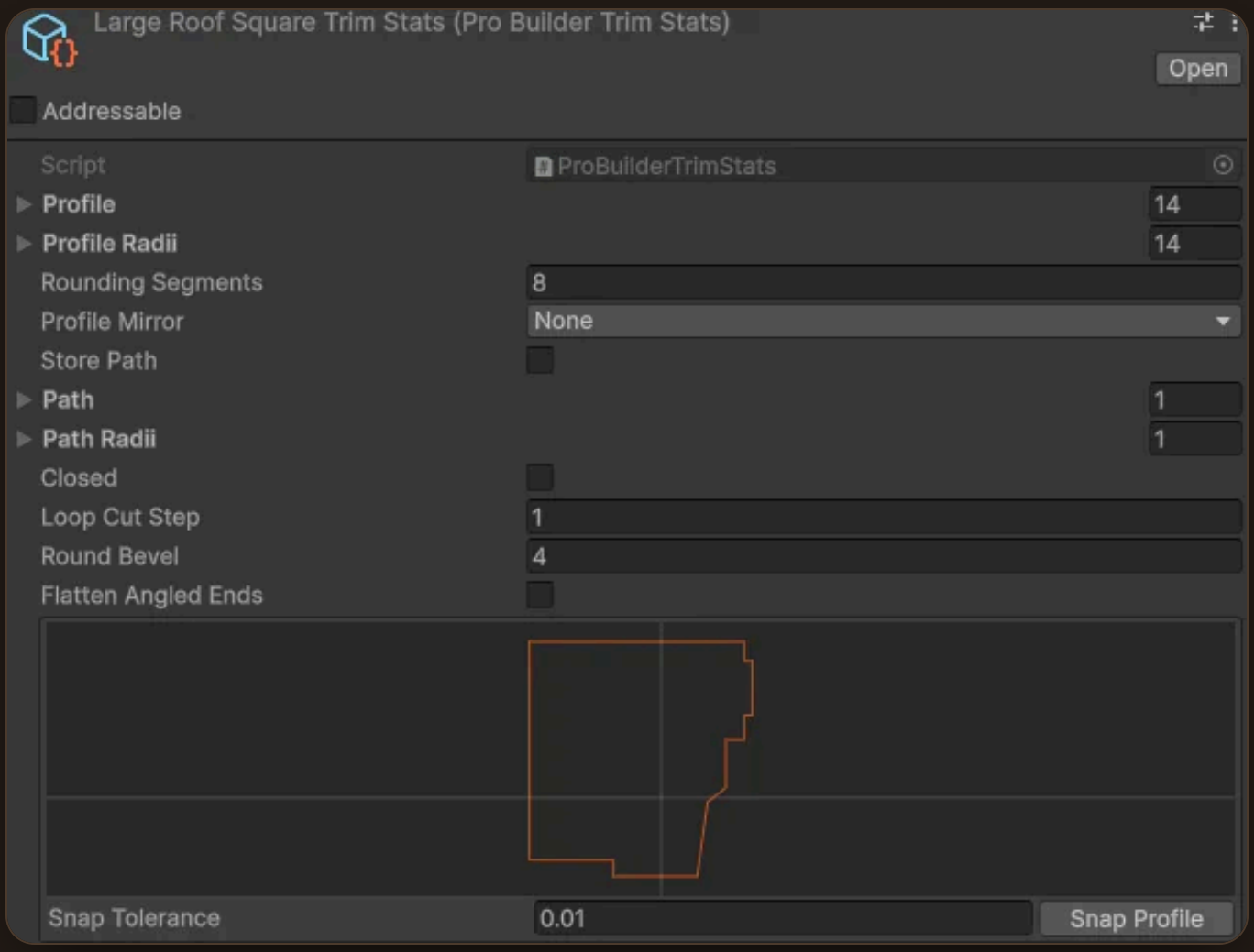


A trim following a path, with its profile edited at the start.

1. Create an empty GameObject and add **Add Component** › **tsitski ProBuilder Tools** › **ProBuilder Trim**.
2. Click the dot past the path's end to extend it, and place points where the trim should run.

3 Shape the profile: its points (in pink) are drawn at the start of the path. X points to the right of the path, Y points up, and (0, 0) sits on the path.

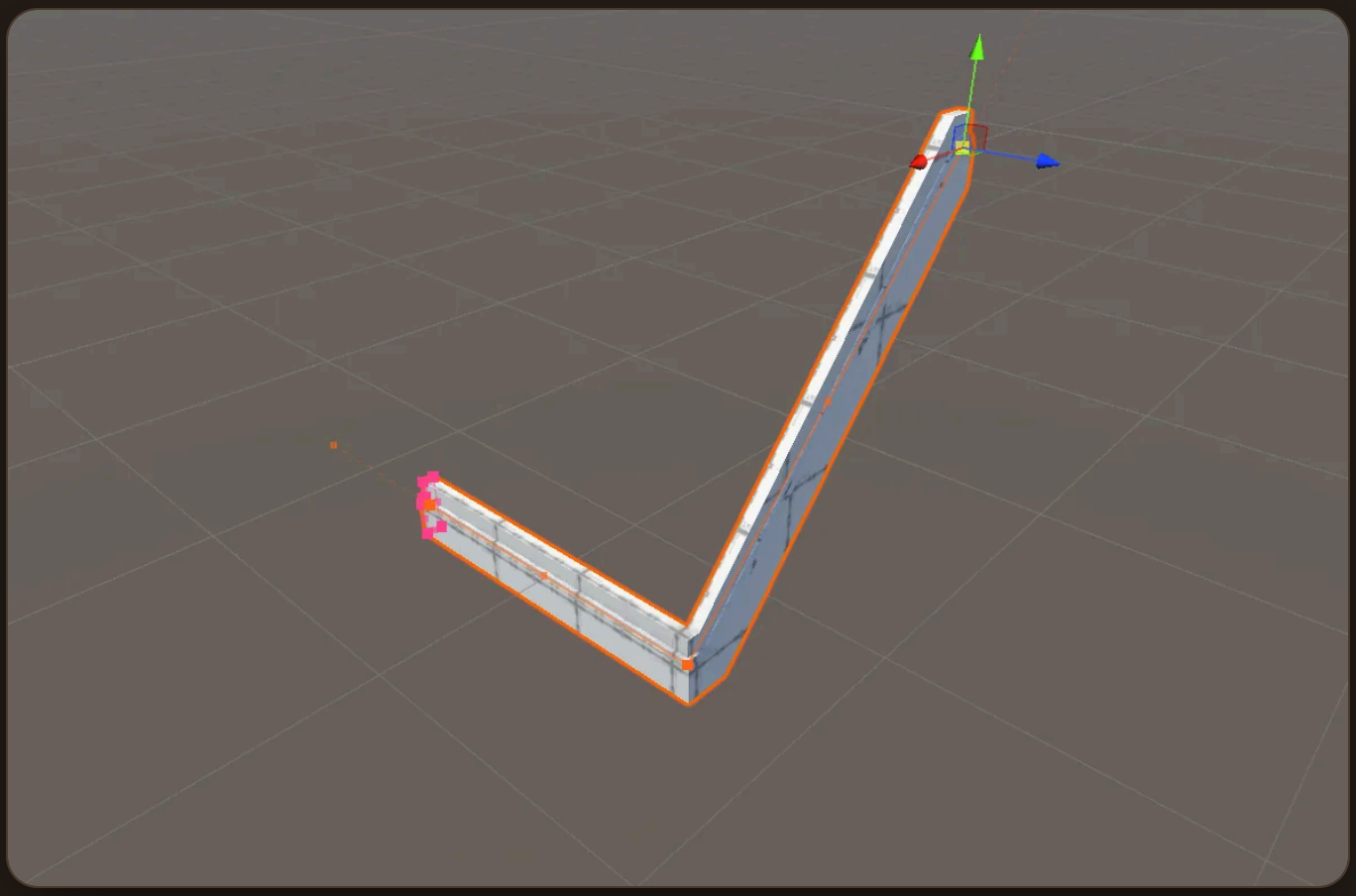
Profile display



The profile display and snap tools.

The inspector draws the finished profile in 2D. Under it, **Snap Profile** lines up profile points that are within **Snap Tolerance** of each other on X or Y, to tidy up hand-placed points.

Sloped paths



The turn is made level, then the slope continues.

Where a path both turns and climbs at a sharp corner, the turn is made level at the corner's height with a clean mitre, and the slope starts just after it, so the profile never twists. Rounded corners stay smooth curves. **Flatten Angled Ends** does the same at the start and end of a sloped path so the end caps stand upright.

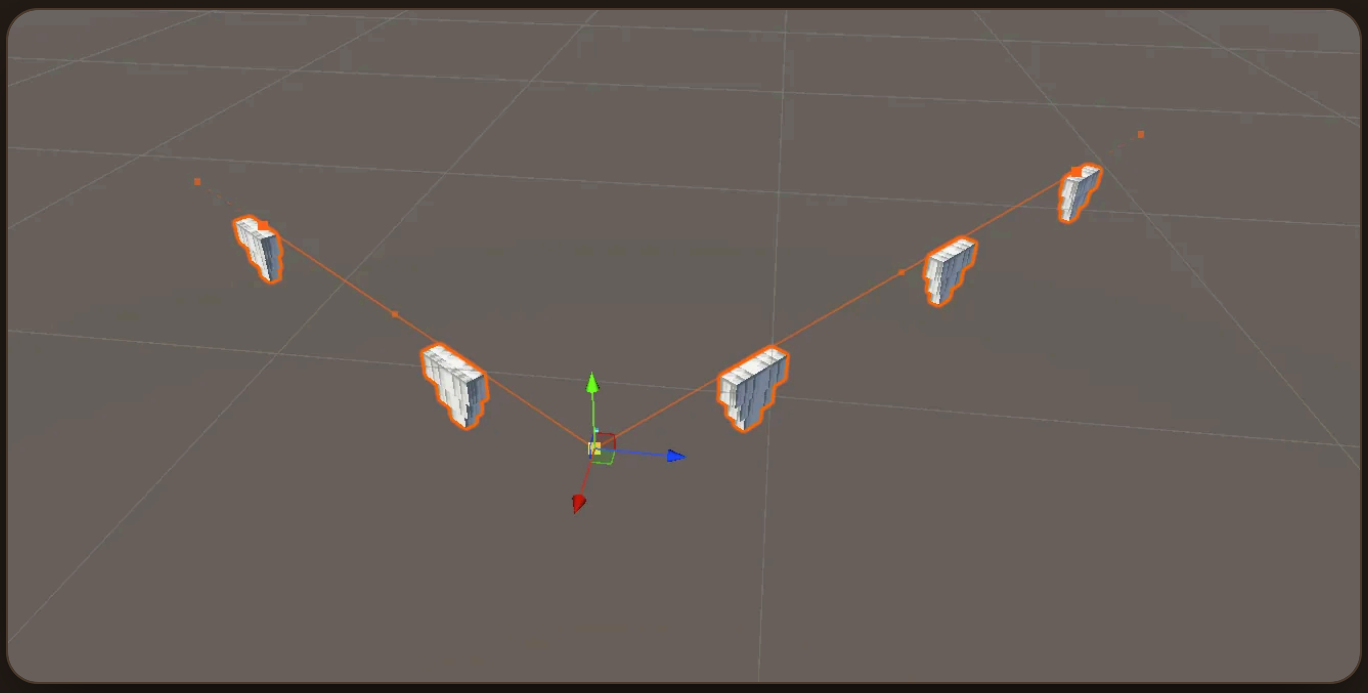
PROPERTY	TYPE	WHAT IT DOES
Stats	Trim Stats	Shared profile and settings. With **Store Path** on, the asset holds the path too.
Profile / Profile Radii	List of Vector2 / float	The cross-section and the rounding of each of its corners.
Path / Path Radii	List of Vector3 / float	The path in local space, and the rounding of each corner.
Closed	bool	Joins the last point back to the first.
Cap Ends	bool	Closes the ends of an open trim.
Smoothing Angle	0–180	Faces meeting at less than this angle are shaded smooth.
Loop Cut Step	float	Distance between extra loop cuts along each segment. Below 0.1 turns them off.
Round Bevel	int	Segments used to round every path corner. 0 keeps sharp mitred corners.
Rounding Segments	int	How smooth rounded path and profile corners are.
Profile Mirror	enum	None, Across Y, Across X or Both. Draw half the profile and have it mirrored to complete the shape.
Flatten Angled Ends	bool	Runs the trim level for a short way at sloped ends so the caps are upright.

TIP

To cut a trim in two, select some of its path points and run `Tools › tsitski ProBuilder Tools › Modelling › Split Into New Trim`. See Split Into New Trim.

ProBuilder Array

An array copies a source mesh along a path, evenly spaced, into one combined mesh: fence posts, columns, lights, railings.



An array of a source mesh along a path.

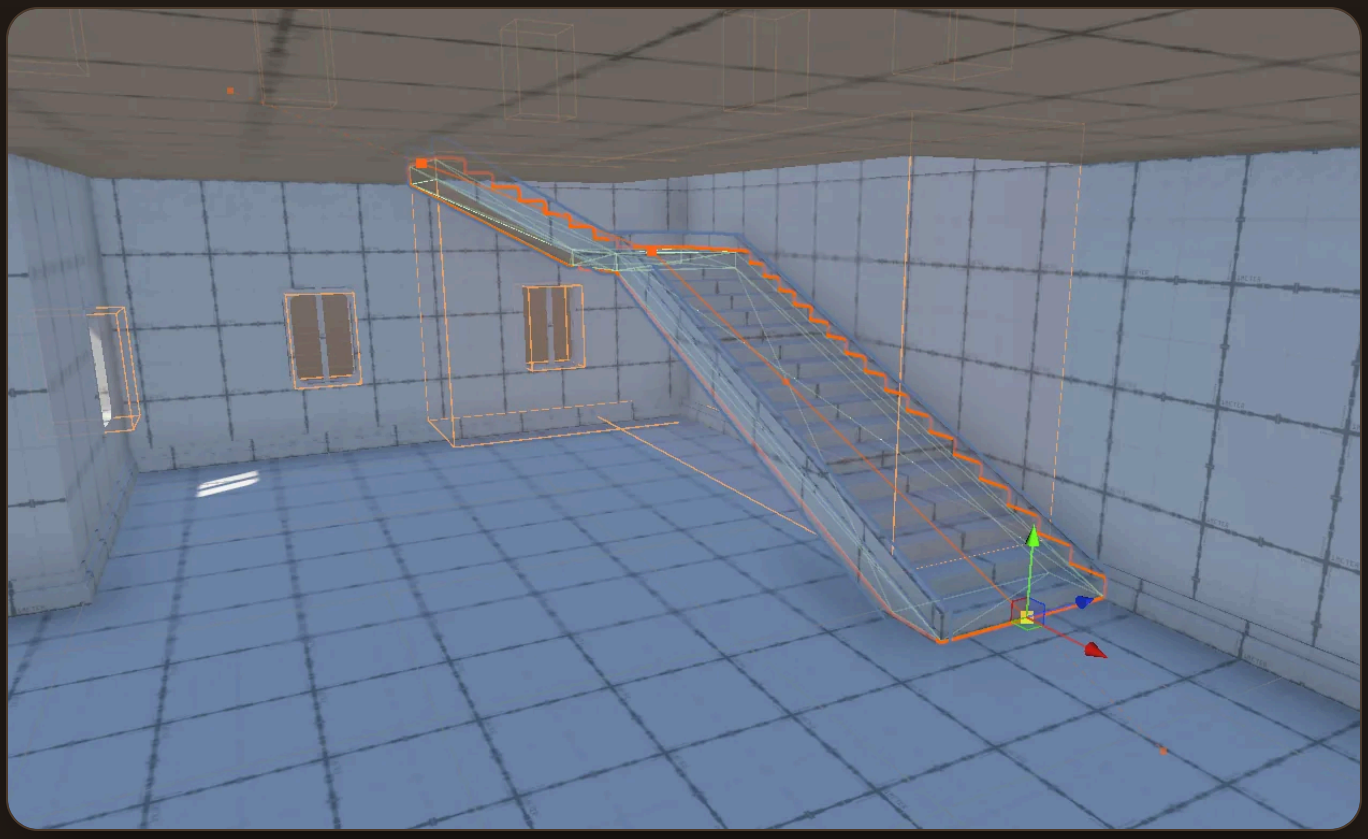
1. Add `Add Component` › `tsitski ProBuilder Tools` › `ProBuilder Array` to an empty GameObject. A small cube source is created as a child (or, on an existing ProBuilder mesh, a copy of it).
2. Model the source with the normal ProBuilder tools. Its +Z points along the path and +Y points up.
3. Edit the path in the Scene view.

PROPERTY	TYPE	WHAT IT DOES
Stats	Array Stats	Shared settings. It can also hold a prefab source, the source mesh itself (**Store Source**) and the path (**Store Path**). Assigning stats with a stored source loads it into the array's source.
Source	ProBuilder Mesh	The mesh that's copied. It's only shown while selected.
Spacing	float	Distance between copies.
Offset	float	Moves the first copy along the path.
Avoid Corners	bool	Keeps copies at least half a Spacing away from corners sharper than 15°.
Space Per Segment	bool	Fills each segment on its own, with a copy on every path point.
Align To Path	bool	Turns each copy to follow the path.
Rounding Segments	int	How smooth rounded path corners are.
Closed	bool	Joins the last point back to the first.

The inspector also has **Select Source**, **Rotate Source 90° Y**, and **Save / Load Source** buttons to move the source mesh in and out of the stats.

ProBuilder Stairs

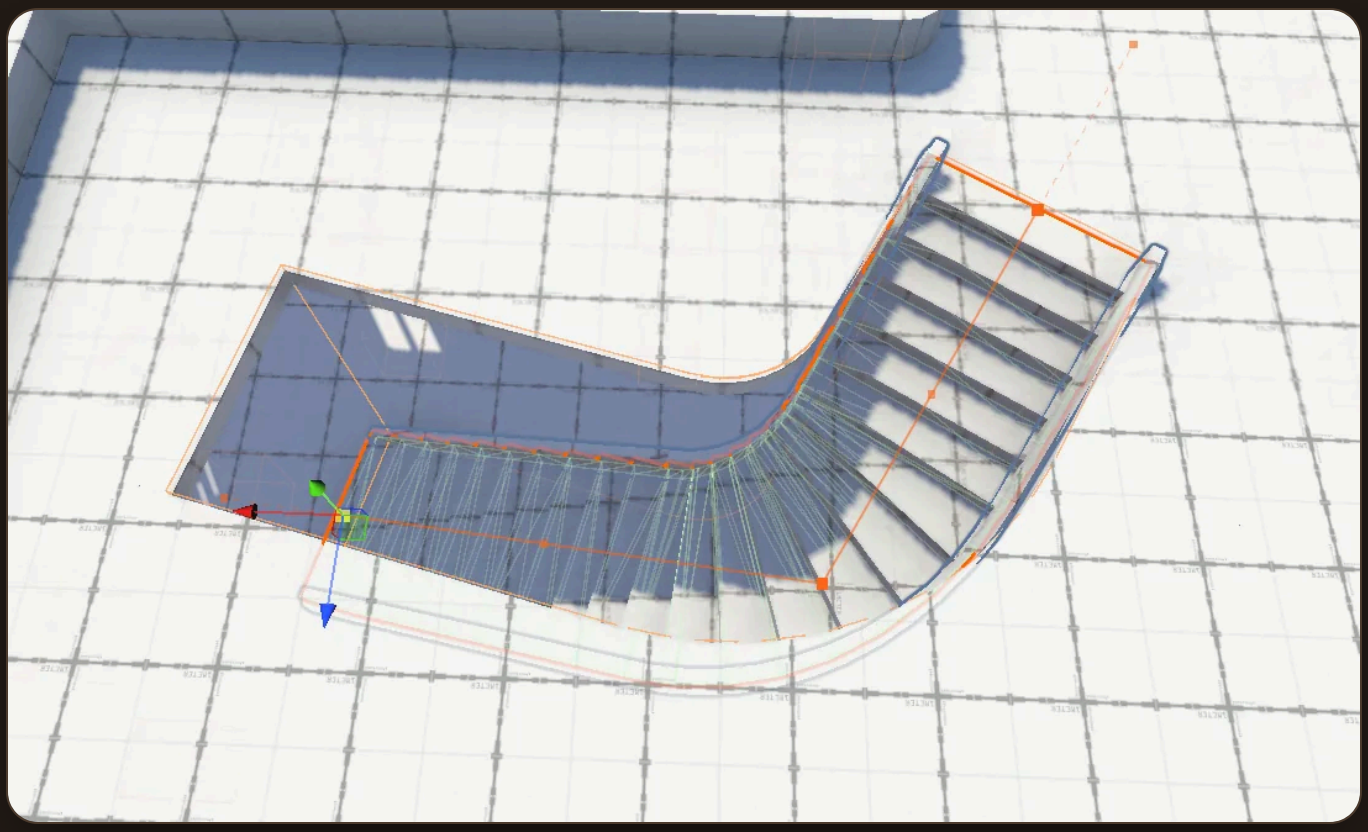
Stairs follow a path. Each pair of points is a flight: the first point is the front edge of the first step, the last point is the back edge of the last step, and every point in between gets a square landing at its height.



Stairs with a landing on the middle point.

1. Add `Add Component` › `tsitski ProBuilder Tools` › `ProBuilder Stairs` to an empty GameObject.
2. Move the path's points to set where the stairs start and end, and add points for landings.
3. Round a corner point to swap its landing for winding steps.

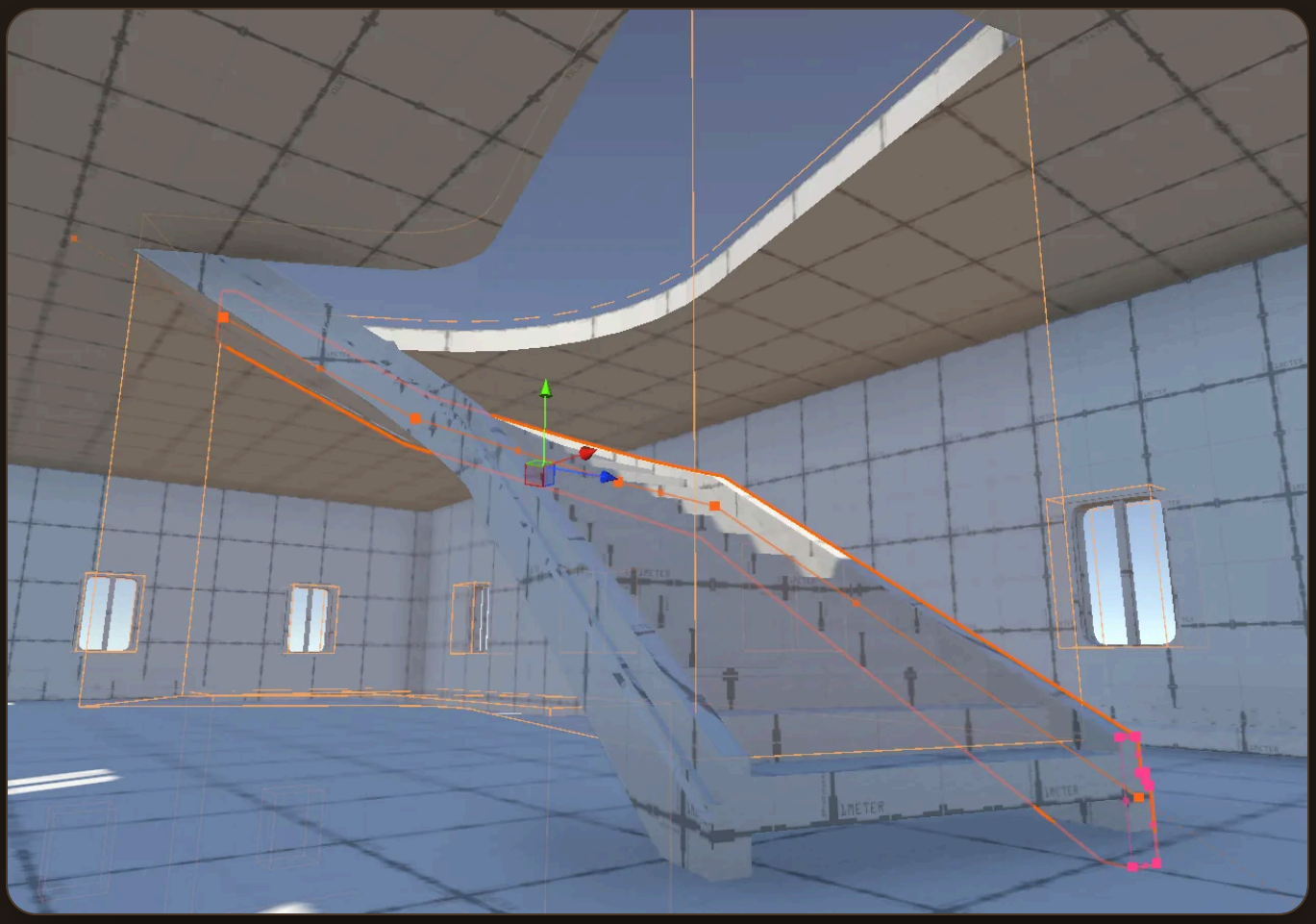
Landings and winding corners



A rounded corner turns the landing into winding steps.

- A 90° turn gets a full square landing, so L-shaped stairs just need three points.
- Rounding a point replaces its landing with winding steps. The radius is the inner edge's radius, so a tiny radius pivots the steps around a point and a bigger one leaves an open well.
- Flights joined by rounded points climb evenly from one landing to the next, so a rounded point's own height is ignored.
- The step count is rounded so every step fits exactly; the inspector shows the total.

Stairwell cutter, trims and collider



Generate Cutter opens a stairwell; side trims follow both edges.

- **Generate Cutter** adds a flat Path cutter following the stairs, so the floor above gets a stairwell cut through it. It only covers the part of the stairs within **Cutter Headroom** of the top, and never cuts the stairs themselves or the floor they start on.
- **Left Trim** and **Right Trim** take Trim Stats and generate a trim along each edge of the stairs and landings, curving round winding corners. The profile's X points away from the stairs on both sides, so one asset makes a mirrored pair.
- **Smooth Collider** gives the Mesh Collider a ramp instead of steps, so characters walk up smoothly. The visible stairs keep their steps.

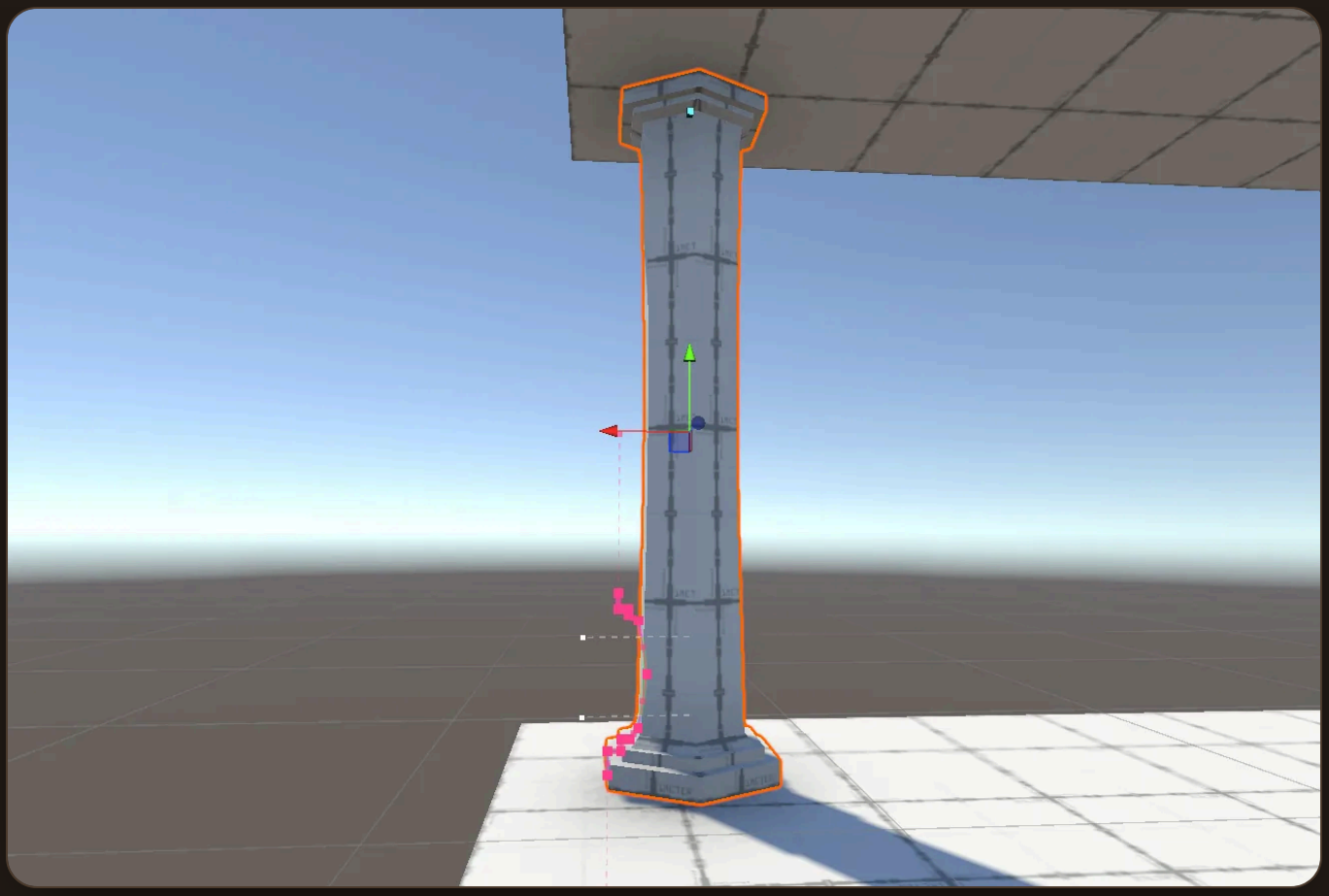
PROPERTY	TYPE	WHAT IT DOES
Stats	Stairs Stats	Shared width, step height, solid, thickness, cutter, trim and collider settings.
Path / Path Radii	List	The flights and corner rounding.
Width	float	Width of the stairs and landings.
Step Height	float	Target riser height.
Solid	bool	Fills down to the lowest path point. Off gives a sloped underside instead.
Thickness	float	Underside and landing thickness when Solid is off.
Generate Cutter	bool	Adds the stairwell cutter.
Cutter Headroom	float	How far below the top of the stairs the stairwell cutter reaches.
Left Trim / Right Trim	Trim Stats	Generates side trims.
Smooth Collider	bool	Ramp collider instead of steps.

NOTE

The cutter and trims are generated children of the stairs and rebuilt with them, so don't edit them by hand.

ProBuilder Pillar

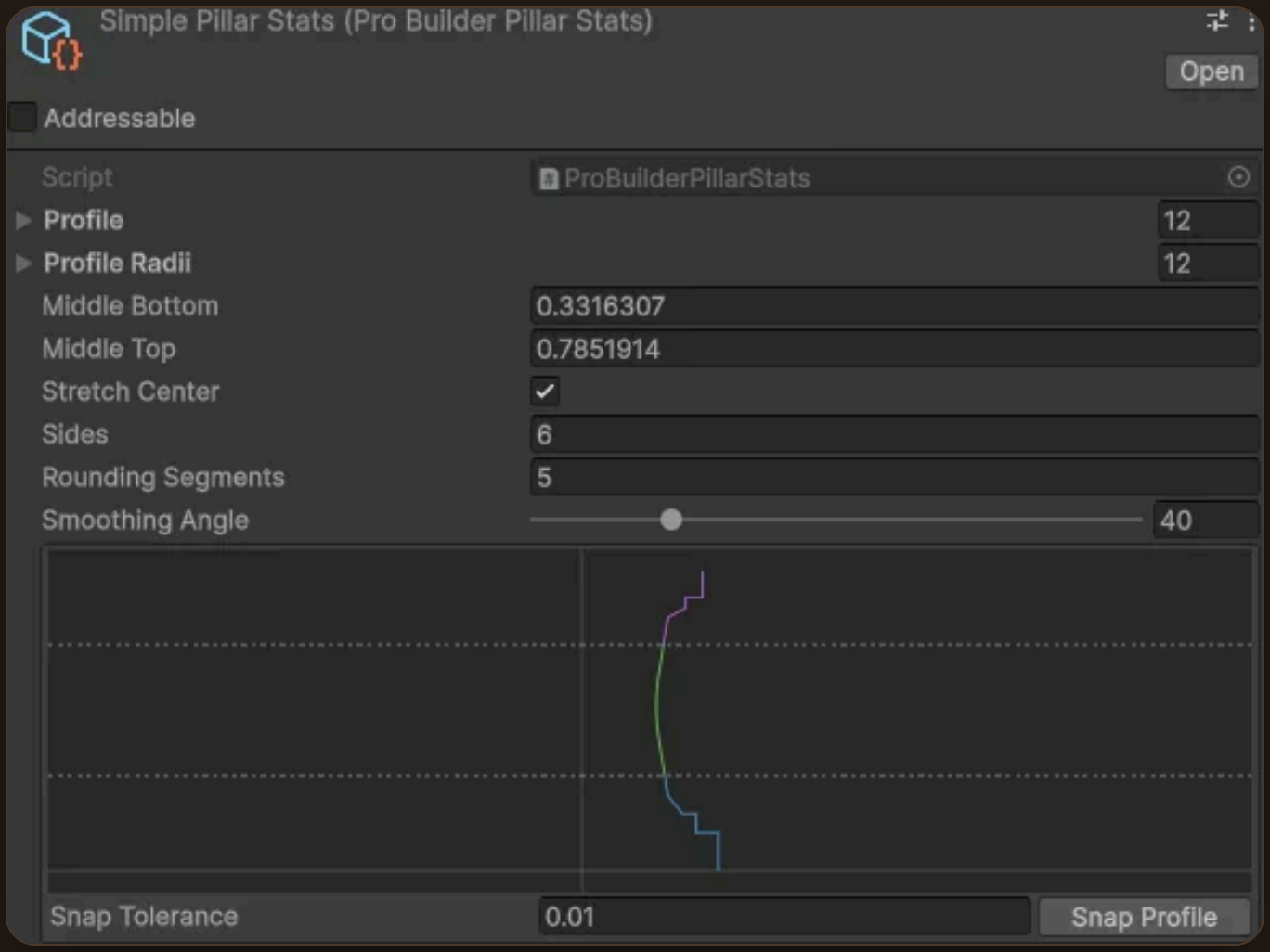
A pillar spins a profile around its vertical axis. The profile is split into three parts: a **bottom** (base) and **top** (capital) that keep their size, and a **middle** (shaft) that repeats, or stretches, to fill the pillar's height.



A 12-sided pillar and a square pillar from the same profile.

1. Add `Add Component` › `tsitski ProBuilder Tools` › `ProBuilder Pillar` to an empty GameObject.
2. Edit the profile's pink points on the pillar's +X side. X is the distance from the axis to the middle of a flat side, Y is height.
3. Drag the two white handles to set where the middle part starts and ends.
4. Drag the cyan handle on top of the pillar to set its height.

The profile display in the inspector colours the bottom blue, the middle green and the top purple, with the two split heights as dotted lines, and has the same Snap Profile tools as trims.

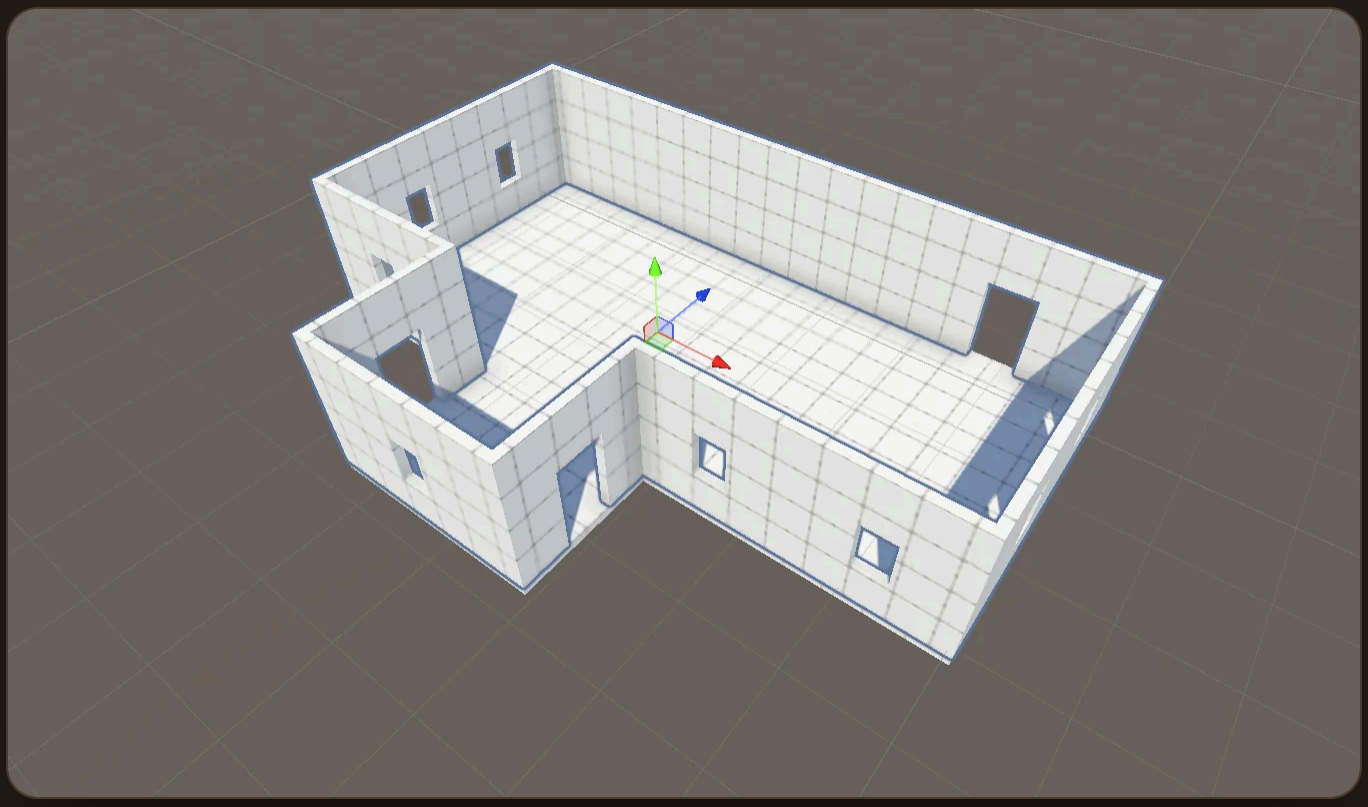


Bottom, middle and top parts of the profile.

PROPERTY	TYPE	WHAT IT DOES
Stats	Pillar Stats	Shared profile and settings.
Height	float	Total height of the pillar.
Profile / Profile Radii	List	The open outline that's spun.
Middle Bottom / Middle Top	float	Heights in the profile where the middle part starts and ends.
Stretch Center	bool	Stretches the middle part once instead of repeating it.
Sides	int	Number of sides. 4 makes a square pillar with its faces along X and Z.
Rounding Segments	int	How smooth rounded profile corners are.
Smoothing Angle	0–180	Sides and profile bends under this angle are shaded smooth, so square pillars stay crisp.

Walls From Floor Plan

Generates walls around the outline of a floor's top faces, and keeps them up to date as you edit the floor.



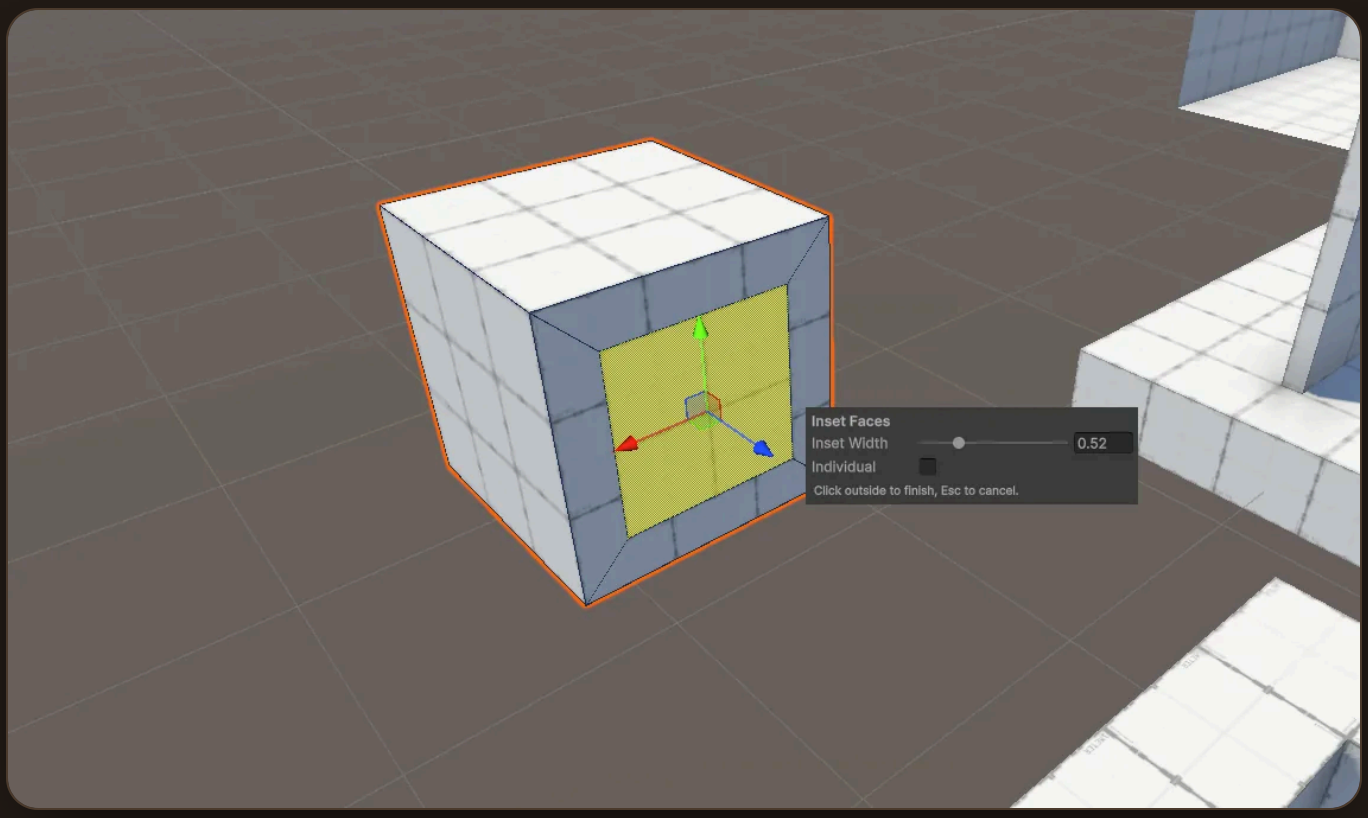
Walls following the outline of a floor.

1. Select a ProBuilder floor mesh.
2. Run `Tools` › `tsitski ProBuilder Tools` › `Generate` › `Walls From Floor Plan`. A `<floor> Walls` child with a **ProBuilder Floor Plan** component is created.

PROPERTY	TYPE	WHAT IT DOES
Floor	ProBuilder Mesh	The floor the walls follow.
Height	float	Wall height.
Thickness	float	Wall thickness.
Alignment	enum	Inside, Center or Outside: which side of the floor's edge the walls sit on.
Walls Around Holes	bool	Also builds walls around holes in the floor.
Square Corners	bool	A separate block at each corner instead of a mitred joint.

Bevel and Inset Faces

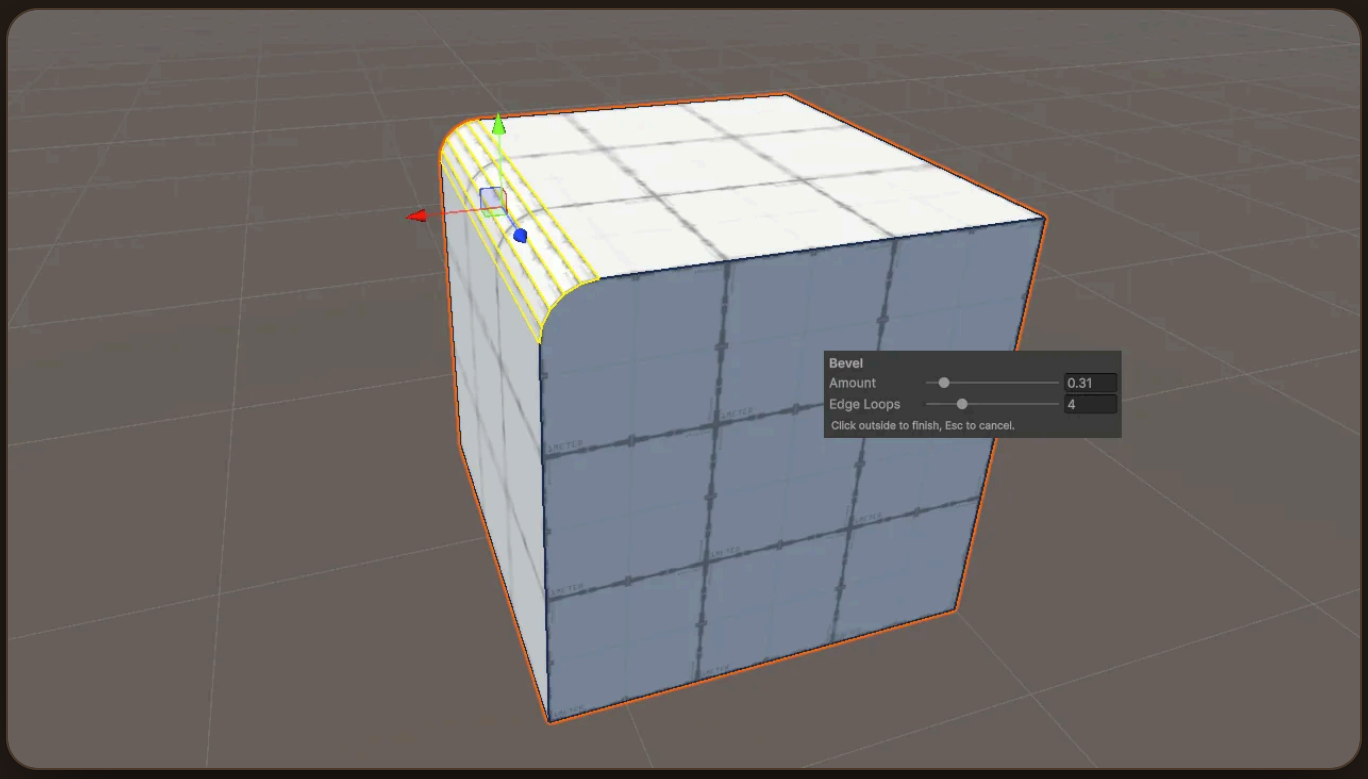
Bevel and Inset Faces apply straight away and open a small settings window at your cursor, like Blender's redo panel. Drag its sliders and the result updates live.



The settings window opens at your cursor.

- Click outside the window, or press `Enter`, to keep the result.
- `Esc` cancels and puts the mesh back.
- The whole operation is one undo step, and the window remembers your last values.

Bevel

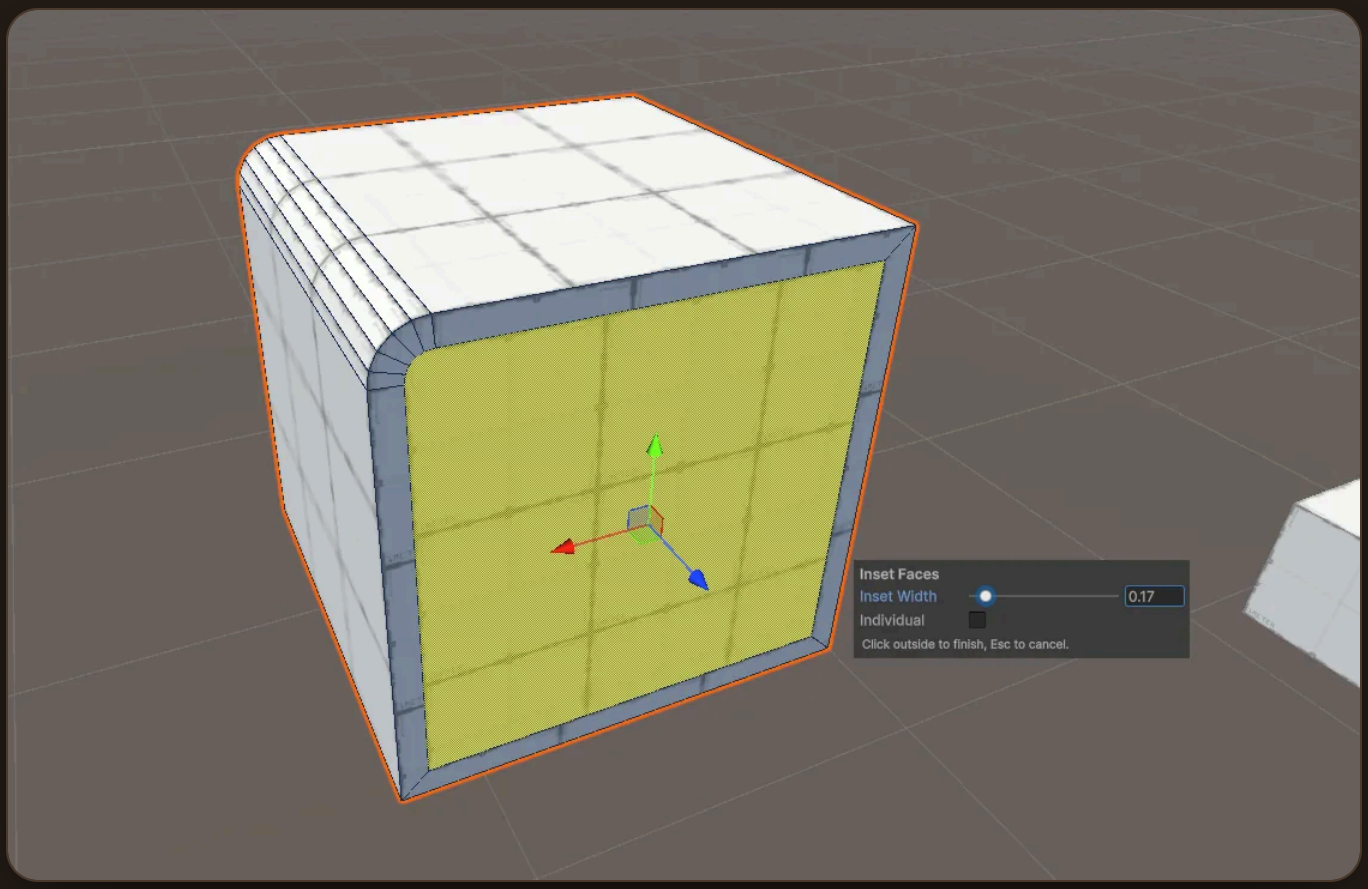


A rounded bevel with Edge Loops.

Bevels the selected edges in edge mode, or the outer edges of the selected faces in face mode. Run `Tools` › `tsitski ProBuilder Tools` › `Modelling` › `Bevel` or press `Alt` + `B`.

- **Amount** — how far the bevel reaches. The slider stops at the longest bevel the surrounding edges allow.
- **Edge Loops** — 0 gives a flat chamfer; more loops round the bevel off into true circular arcs, and the faces around it are updated so there are no gaps. Corners where three bevelled edges meet, like a cube's, get a rounded patch made of quads, like Blender's.

Inset Faces



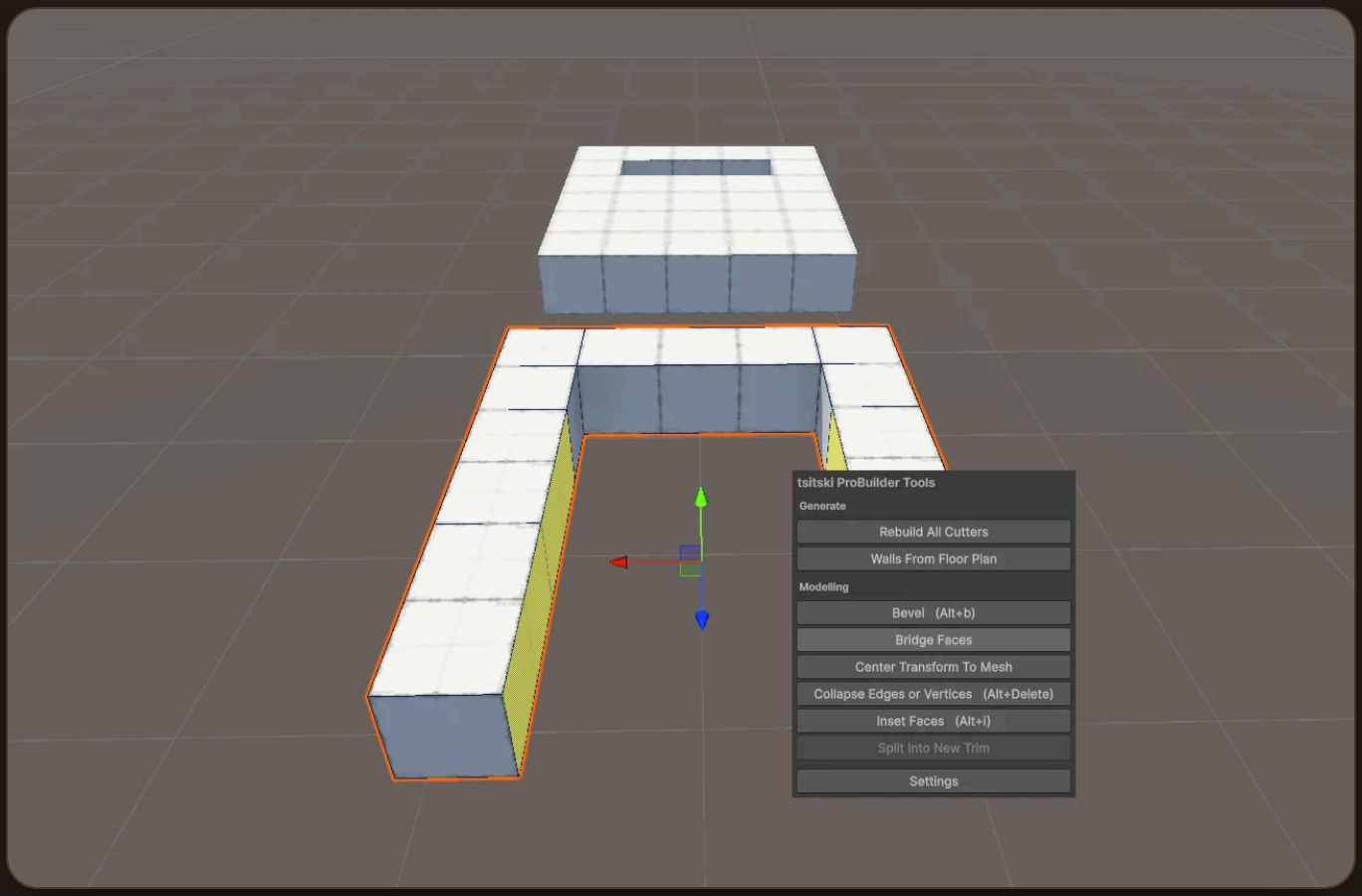
Inset Faces on a group of faces.

Insets the selected faces in face mode. Run `Tools` › `tsitski ProBuilder Tools` › `Modelling` › `Inset Faces` or press `Alt` + `I`.

- **Inset Width** — how far the new edge sits in from the old one.
- **Individual** — insets each face on its own instead of the selection as one region.

Bridge Faces

Connects two faces, or two groups of faces, with a ring of quads, then deletes the faces you selected. It works like Blender's Bridge Edge Loops, lining the outlines up so the result doesn't twist. Faces that face each other make a tube between them; faces that face away from each other, like the front and back of a wall, make a tunnel through it with its faces pointing inward.



Two faces bridged into a tunnel.

1. Select the mesh and switch ProBuilder to face mode.
2. Select two faces, or two separate groups of faces, on the same mesh.
3. Run `Tools` › `tsitski ProBuilder Tools` › `Modelling` › `Bridge Faces`, or pick it from the `Alt` + `X` popup.

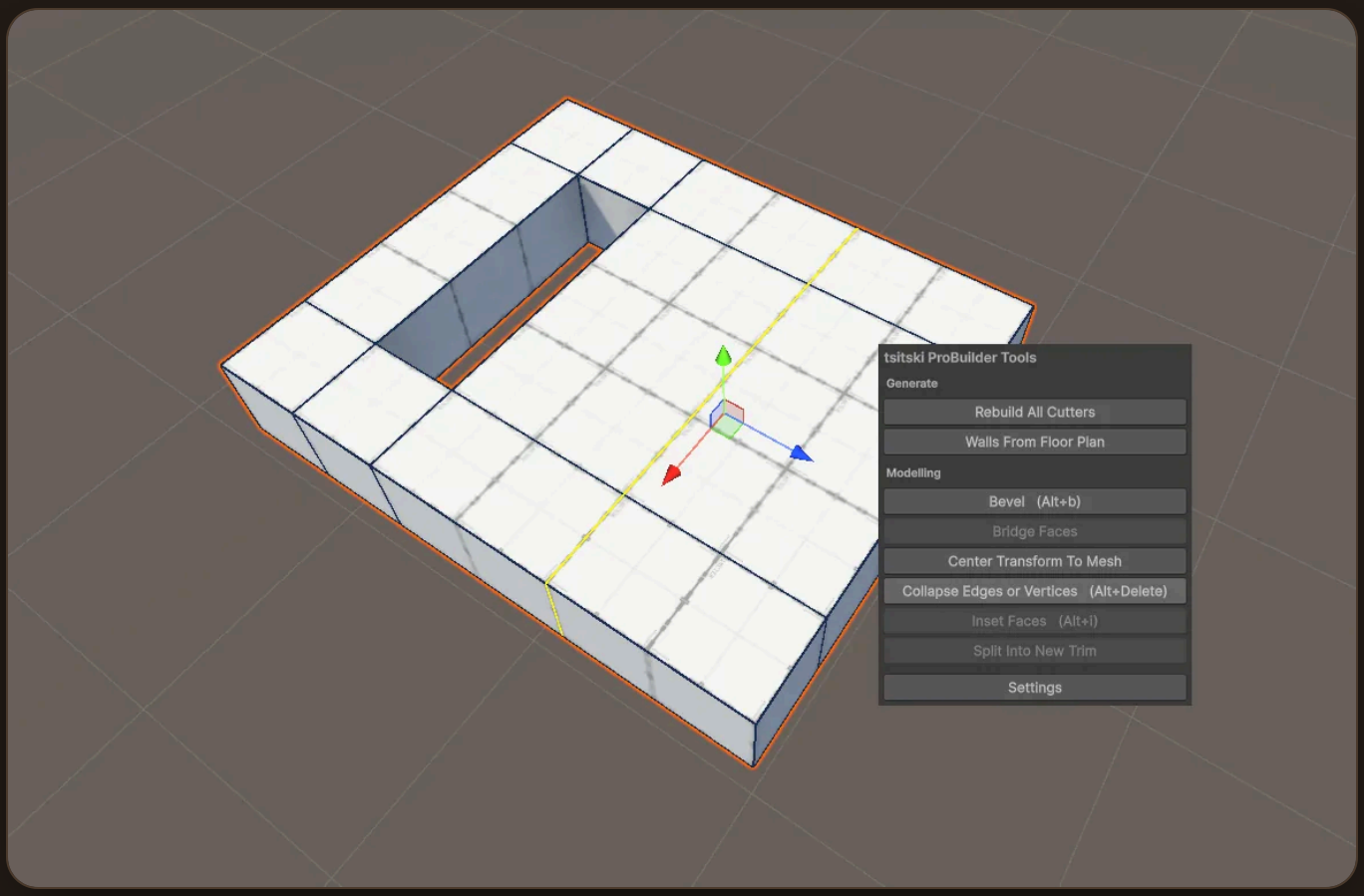
<details>
<summary>What can be bridged</summary>

- Exactly two groups. Faces that share an edge count as one group.
- Each group's outline has to be one loop, with no holes.
- Both outlines need the same number of edges.
- The two groups can't share any vertices.

If something doesn't fit, nothing changes and the Console says why.
</details>

Collapse Edges or Vertices

Cleans up flat geometry. In edge mode it merges the flat faces on both sides of each selected edge. In vertex mode it merges the flat faces around each selected vertex, and in face mode it does the same for every vertex of the selected faces. Vertices left sitting in a straight line along an edge are removed.



Extra edges on a flat wall collapsed into one face.

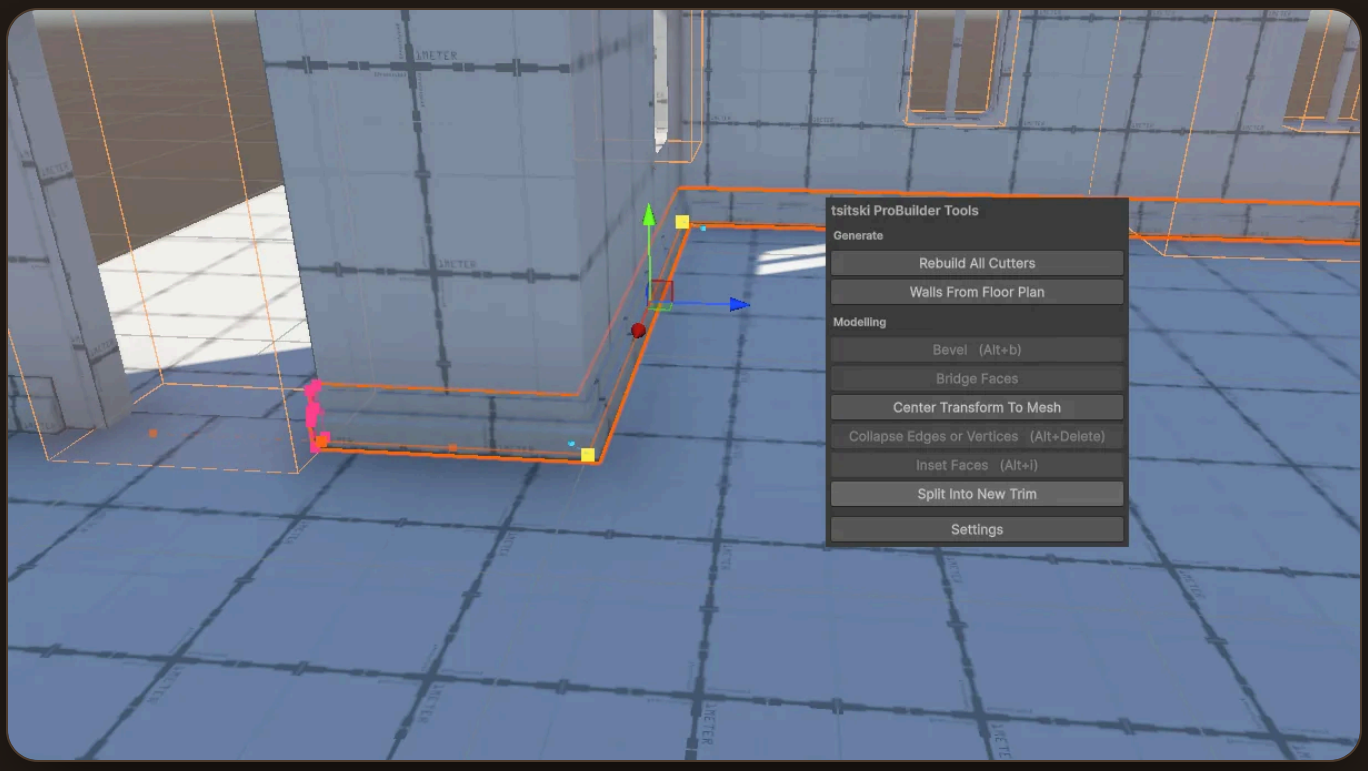
Press `Alt` + `Delete`, or run
Tools › tsitski ProBuilder Tools › Modelling › Collapse Edges or Vertices.

NOTE

Only faces that lie flat against each other are merged. Corners are skipped, and the Console says which.

Split Into New Trim

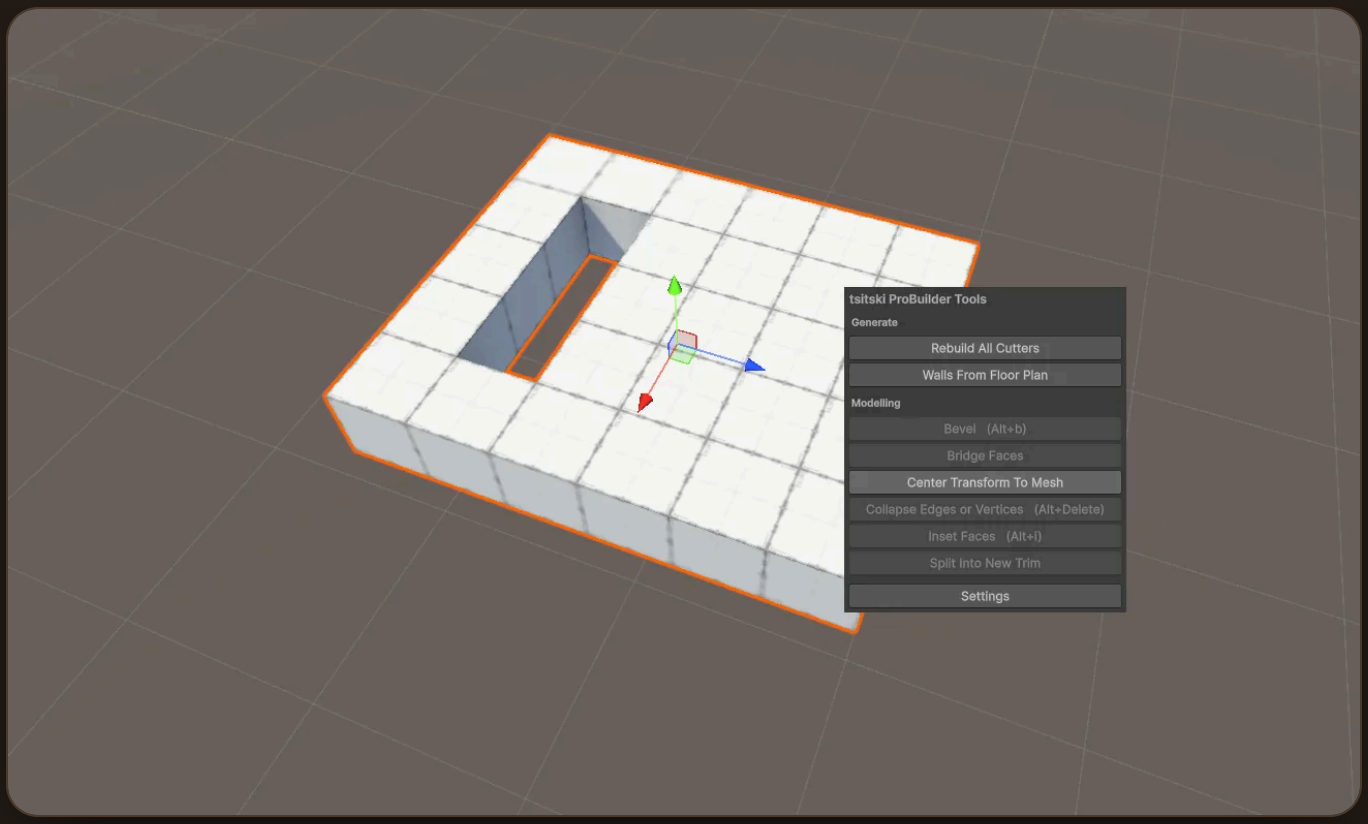
Select path points on a trim and run `Tools` › `tsitski ProBuilder Tools` › `Modelling` › `Split Into New Trim`. Every stretch between two neighbouring selected points becomes a new trim with the same settings, and the pieces stay joined. The new trims are selected, and the split is one undo step.



The selected stretch becomes its own trim.

Center Transform To Mesh

Moves each selected mesh's pivot to the centre of its geometry, snapped to the grid. The mesh and its children stay exactly where they are; only the pivot moves.



The pivot moves to the middle; the mesh stays put.

Select one or more ProBuilder meshes and run `Tools` › `tsitski ProBuilder Tools` › `Modelling` › `Center Transform To Mesh`.

Settings

`Tools` › `tsitski ProBuilder Tools` › `Settings` sets the **Default Mesh Material** given to generated meshes (trims, arrays, stairs, pillars and walls) that don't have one yet. It's stored in `tsitski ProBuilder Tools Settings.asset` in the package folder.

Troubleshooting

▾ My cutter isn't cutting anything

- The mesh has to be a ProBuilder mesh, active, and in the same scene as the cutter.
- The cutter has to actually go into the mesh, not just touch its surface.
- Every axis of the cutter's Size has to be bigger than 0.

- A cutter never cuts meshes in its own prefab or among its own children.
- A cutter that's a child of a ProBuilder mesh skips that mesh unless **Cut Through Parent** is on.
- Still stuck? Run `Tools` › `tsitski ProBuilder Tools` › `Generate` › `Rebuild All Cutters`.

<details>
<summary>The inside walls are missing or doubled</summary>

Check **Create Mesh** and **Generate Faces** on the cutter. When one cutter goes through several meshes, only the one it goes into most deeply gets the walls.
</details>

<details>
<summary>My edits to a trim, array, stairs or pillar mesh disappeared</summary>

Generated meshes are rebuilt from their settings, so direct ProBuilder edits are overwritten. Change the component's settings, path or profile instead.
</details>

<details>
<summary>Bridge Faces says it can't bridge my faces</summary>

The Console message names the problem. See What can be bridged.
</details>

Credits

The cutter's boolean geometry is built on CSG code by Evan Wallace (csg.js), Tomasz Dabrowski (csgjs-cpp) and Karl Henkel (C# port), used under the MIT licence. The full notice ships with the package in `Third-Party Notices.txt`.

Questions, bugs or ideas? Come say hi on Discord.